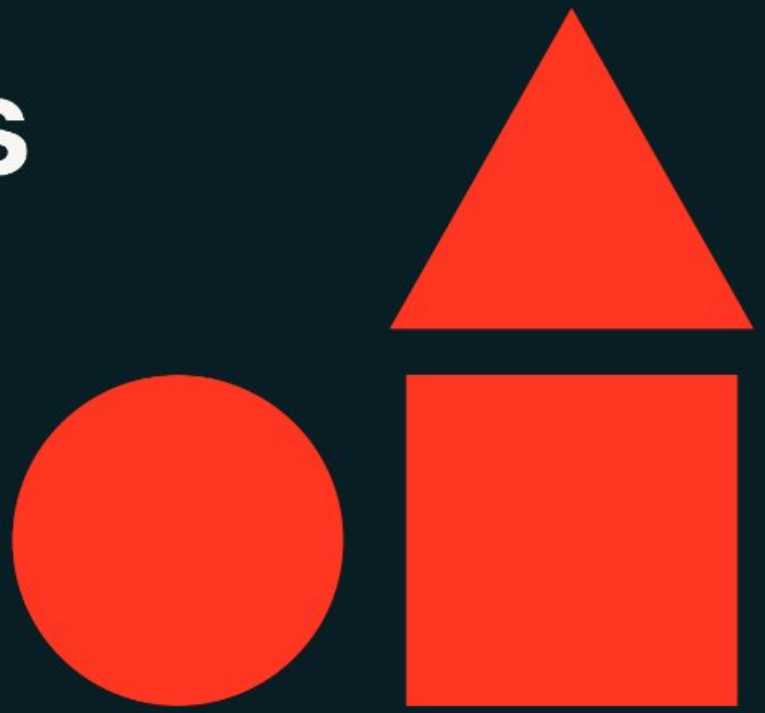
databricks

Build Data Pipelines with Lakeflow Spark Declarative Pipelines



Databricks Academy
May 2026

Agenda

Course Sections
• Introduction to Data Engineering in Databricks
• Lakeflow Spark Declarative Pipeline Fundamentals
• Building Lakeflow Spark Declarative Pipelines



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Course Learning Objectives

- Understand the core concepts and components of Lakeflow Spark Declarative Pipelines (SDP), including the function and differences between streaming tables, materialized views, and temporary views.
- Identify and configure pipeline settings, such as compute, data assets, trigger modes, and advanced options.
- Develop a functional Spark Declarative Pipeline using the new pipeline editor and SQL-based syntax.
- Incorporate data quality expectations into a pipeline to validate and enforce data integrity.
- Analyze event logs and pipeline metrics to understand the full execution and lifecycle of a Spark Declarative Pipeline.
- Design and implement Change Data Capture (CDC) to a pipeline using AUTO CDC INTO to handle slowly changing dimensions (SCD).



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Course Prerequisites (REQUIRED)



Fundamental Knowledge of the Databricks Platform

- Databricks Workspaces
- Apache Spark
- Unity Catalog Tables and the Medallion Architecture
- Unity Catalog



Knowledge of Ingesting Raw Data into a Table

- Familiarity with ingesting raw data source files using the **read_files** SQL function
  - CSV
  - JSON
  - TXT
  - Parquet



Experience Transforming Data with SQL

- Experience in writing **intermediate-level queries** using SQL
- Basic knowledge of **SQL Joins**

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Lab Exercise Environment



Technical Details

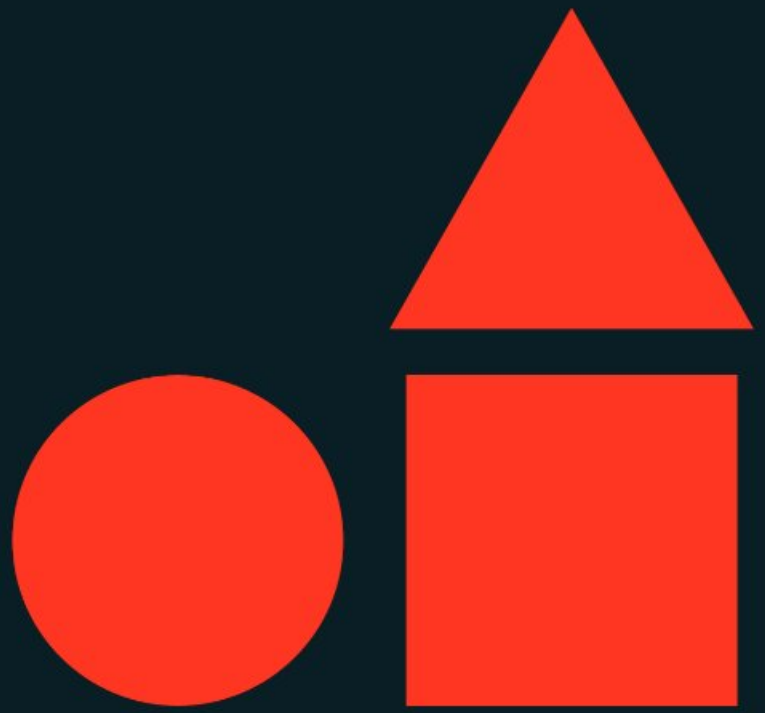
- Your lab environment is provided by Vocareum.
- It will open in a new tab.
- It has been configured with the permissions and resources required to accomplish the tasks outlined in the lab exercise.
- Third party cookies must be enabled in your browser for Vocareum's user experience to work properly.
- Make sure to enable pop ups!



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Introduction to Data Engineering in Databricks



Build Data Pipelines with Lakeflow Spark Declarative Pipelines

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Agenda

Section Overview - Introduction to Data Engineering in Databricks
• Data Engineering in Databricks
• What are Lakeflow Spark Declarative Pipelines?
• Course Setup and Creating a Pipeline
• Course Project Overview



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Section Learning Objectives

- Understand the components of Data Engineering with Lakeflow in Databricks with Connect, Spark Declarative Pipelines, and Jobs.
- Explain how Lakeflow Spark Declarative Pipelines incrementally processes data using Streaming Tables and Materialized Views in batch or streaming Lakeflow Jobs.
- Demonstrate how to create a Lakeflow Spark Declarative Pipelines project in a Databricks Workspace.



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Introduction to Data Engineering in Databricks

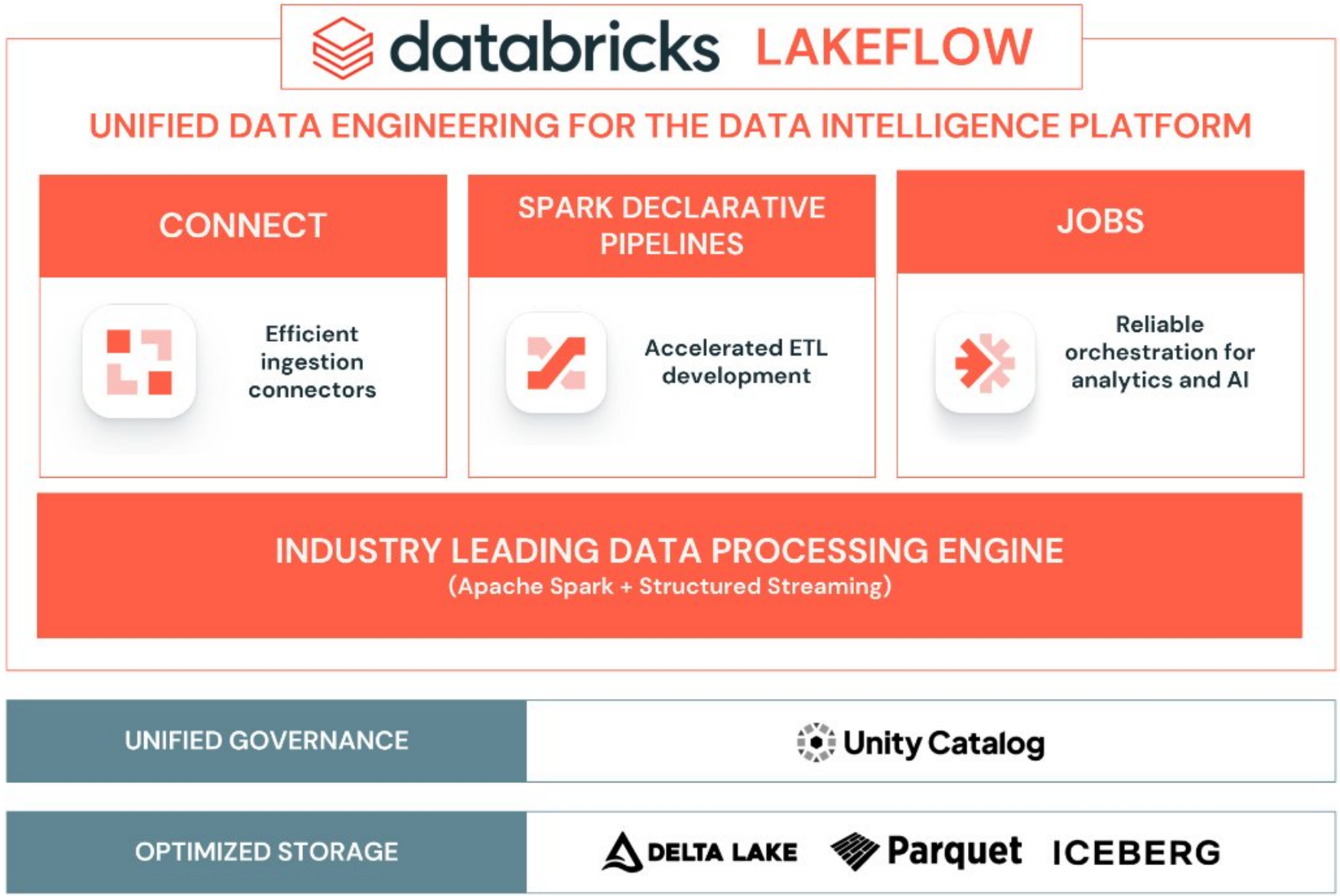
LECTURE

Data Engineering in Databricks



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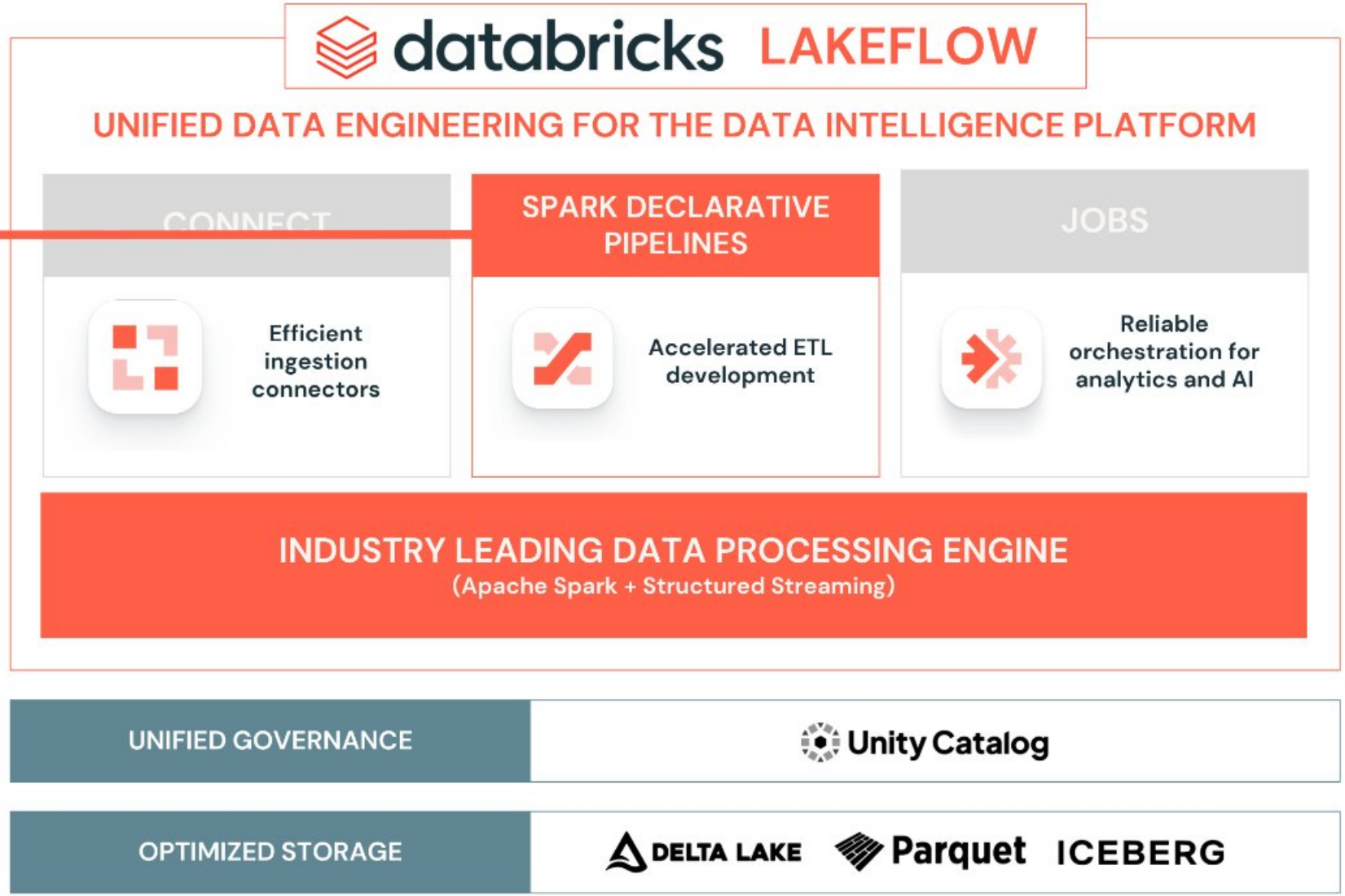
Data Engineering in Databricks



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Data Engineering in Databricks

Simplifying batch and streaming ETL with automated reliability, optimizations and built-in data quality with **Lakeflow Spark Declarative Pipelines (SDP)**!



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Introduction to Data Engineering in Databricks

LECTURE

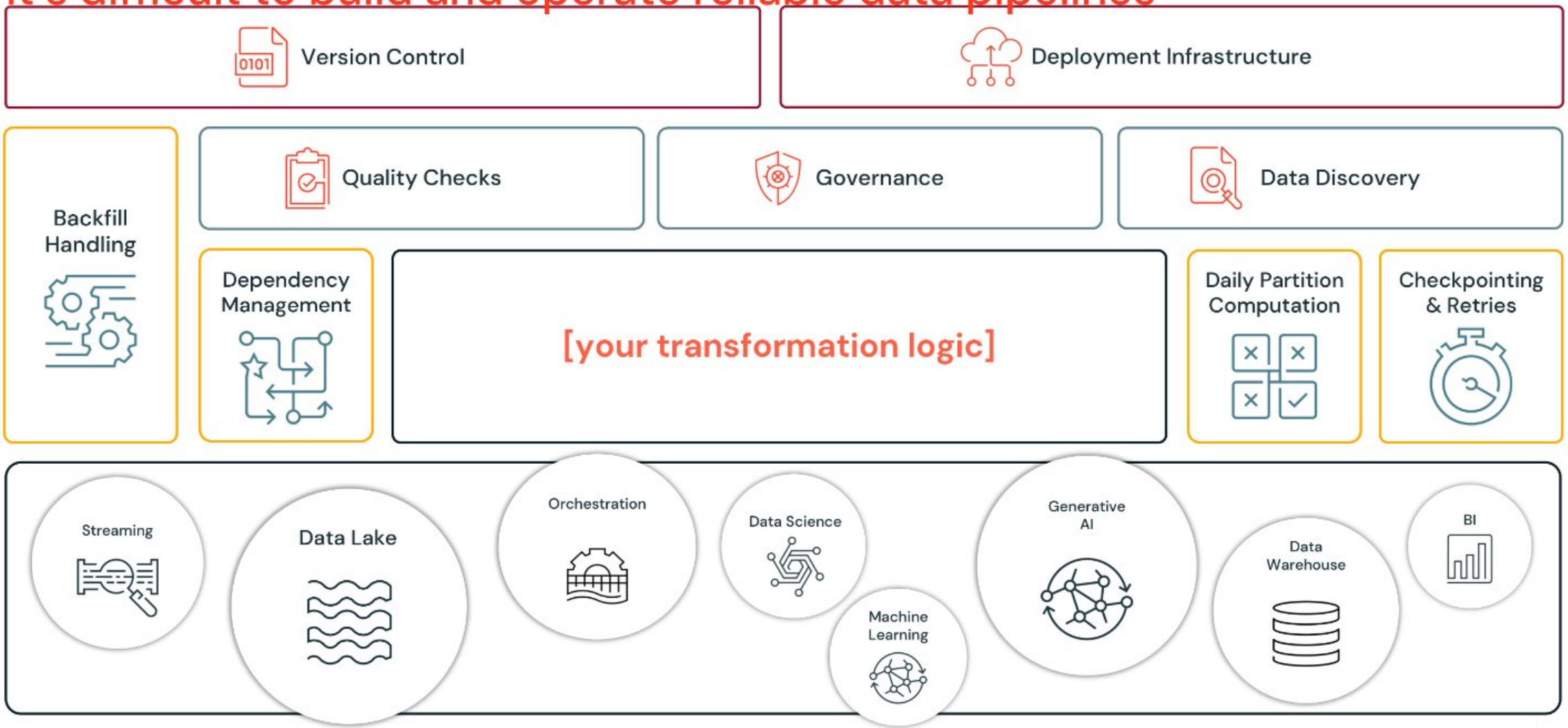
What are Lakeflow Spark Declarative Pipelines?



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What are Spark Declarative Pipelines?

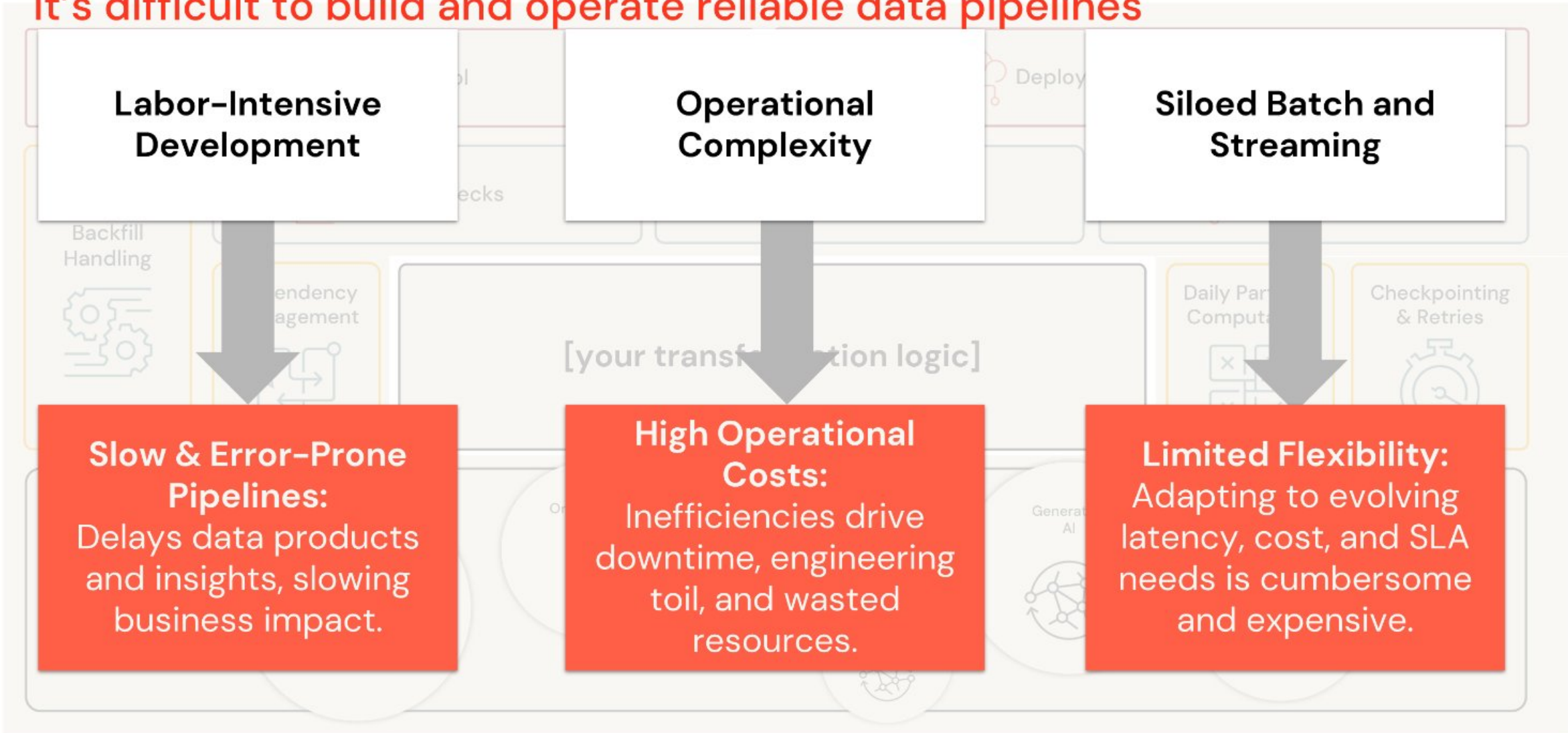
It's difficult to build and operate reliable data pipelines



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What are Spark Declarative Pipelines?

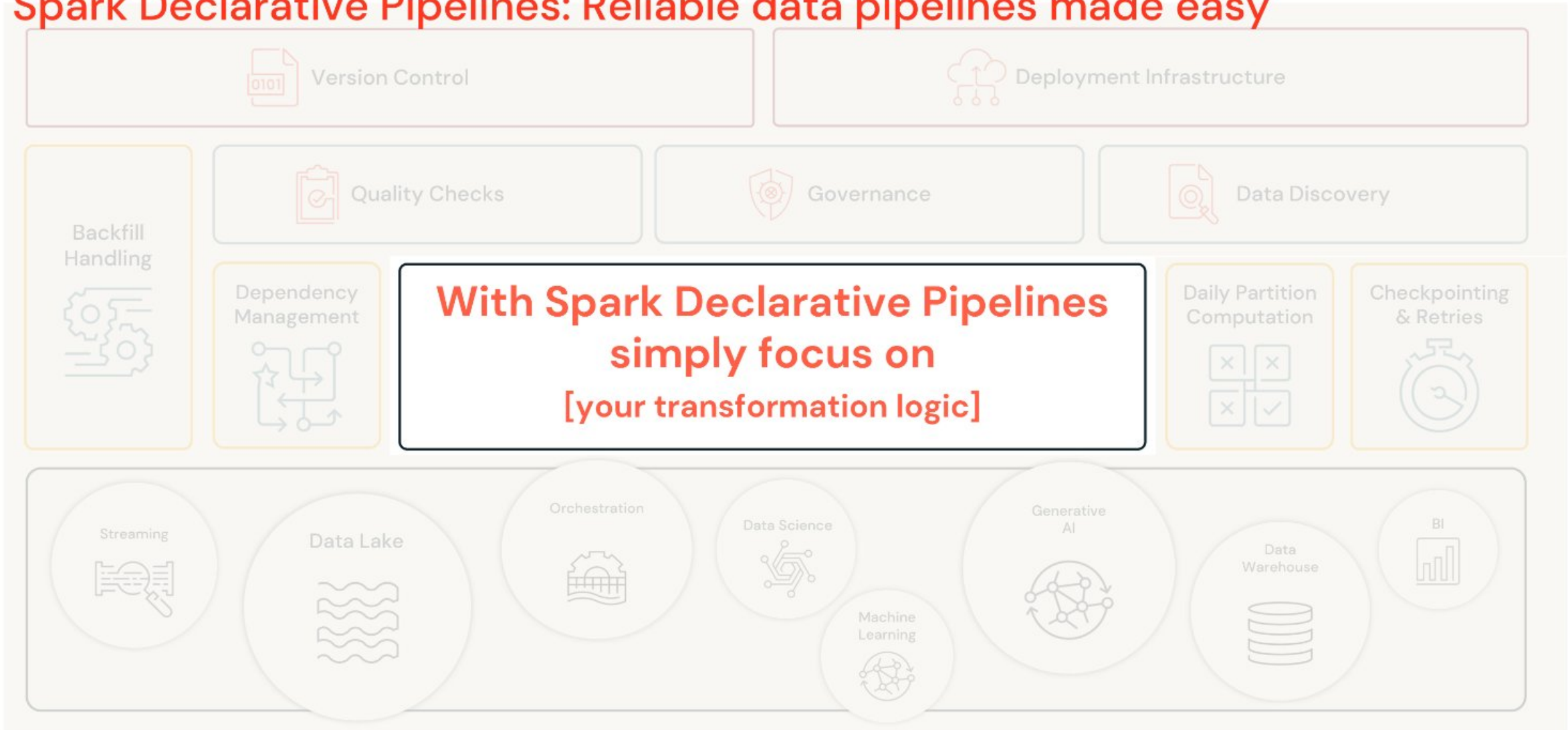
It's difficult to build and operate reliable data pipelines



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What are Spark Declarative Pipelines?

Spark Declarative Pipelines: Reliable data pipelines made easy



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What are Spark Declarative Pipelines?

Introducing Declarative Pipelines: Reliable data pipelines made easy

- **Simplified Pipeline Authoring**

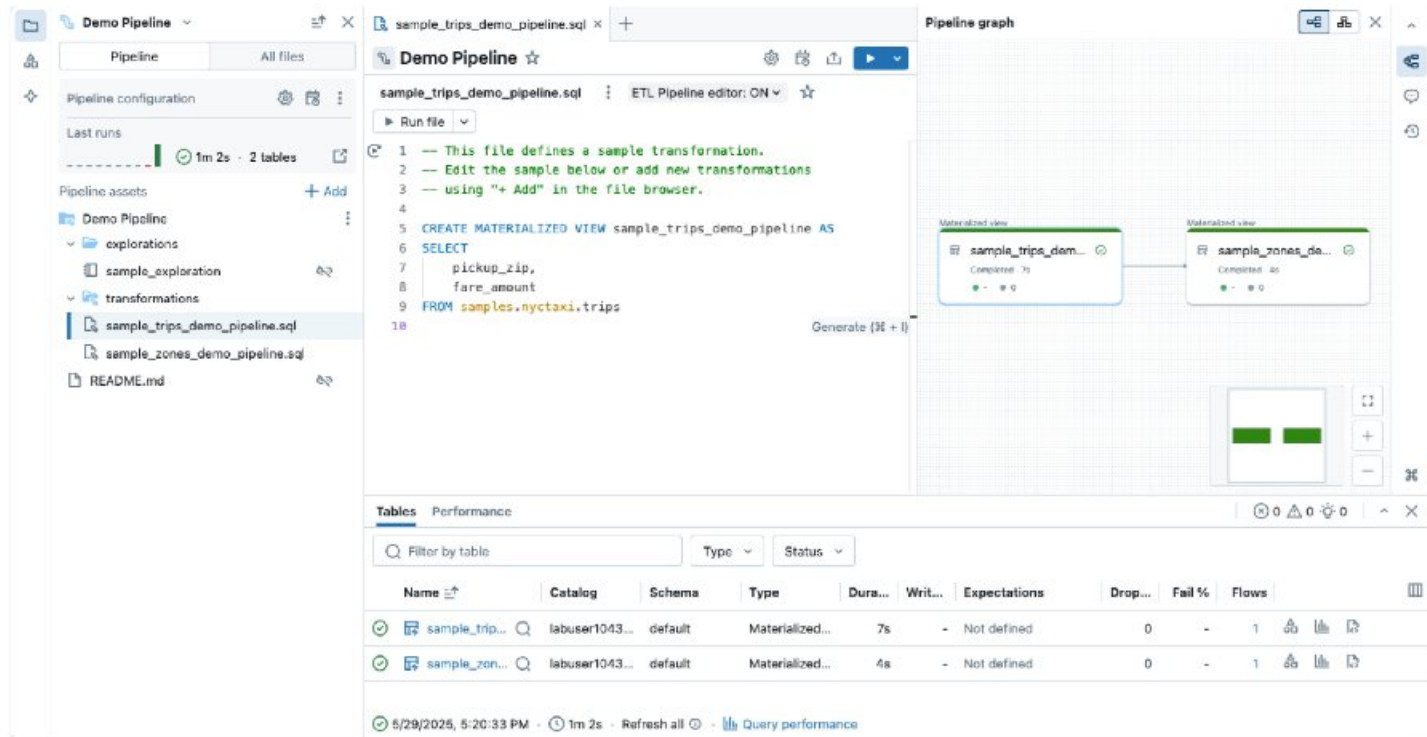
  Easily declare data ingestion and transformations with **SQL or Python,** and let Spark Declarative Pipelines handle the rest!

- **Intelligent Optimization at Scale**

  Automated scaling and recovery improve reliability and reduce maintenance.

- **Unified Batch and Streaming**

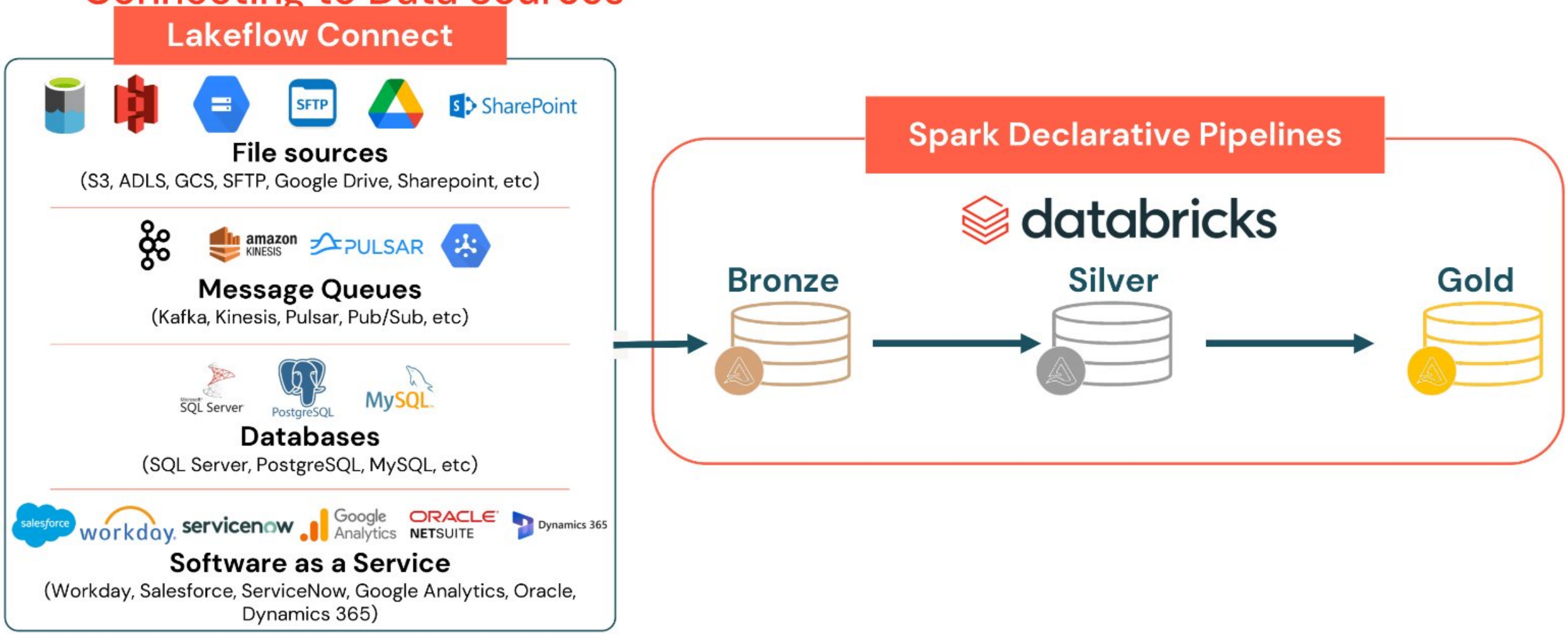
  Pipelines seamlessly adapt to **both near real-time and batch Lakeflow Jobs,** optimizing performance and cost.



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What are Spark Declarative Pipelines?

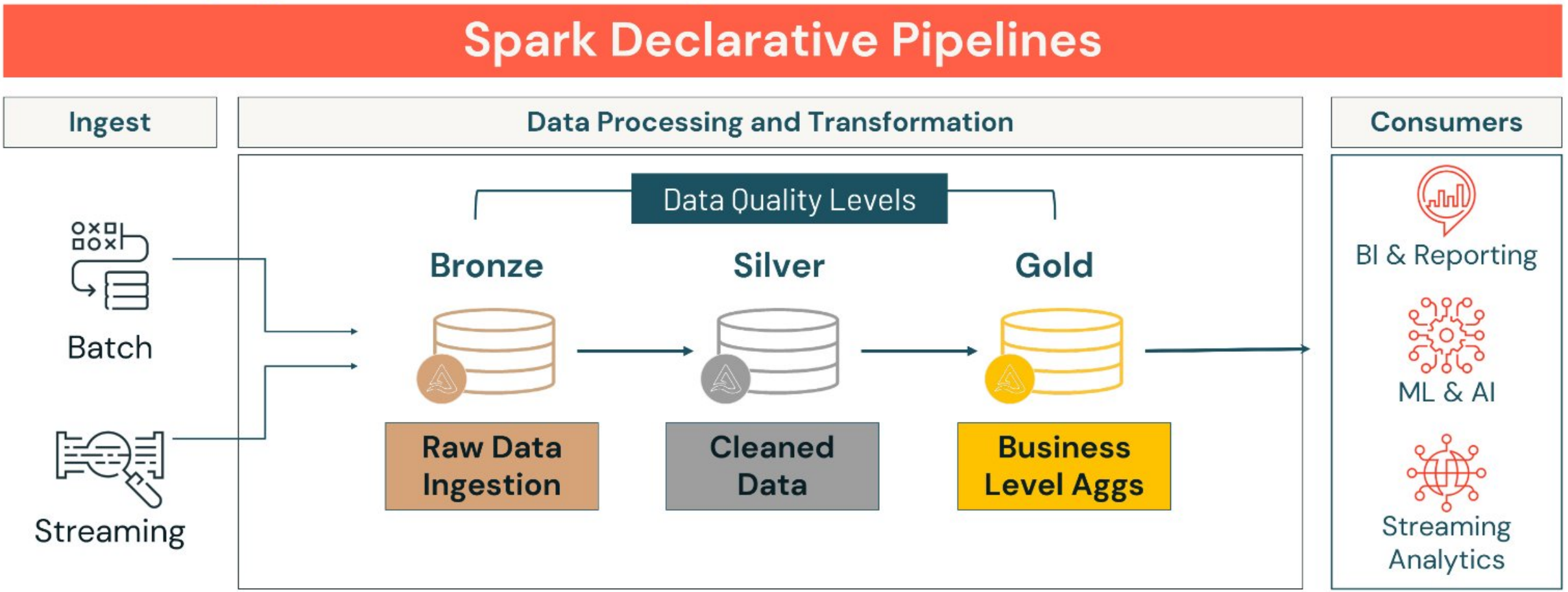
Connecting to Data Sources



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What are Spark Declarative Pipelines?

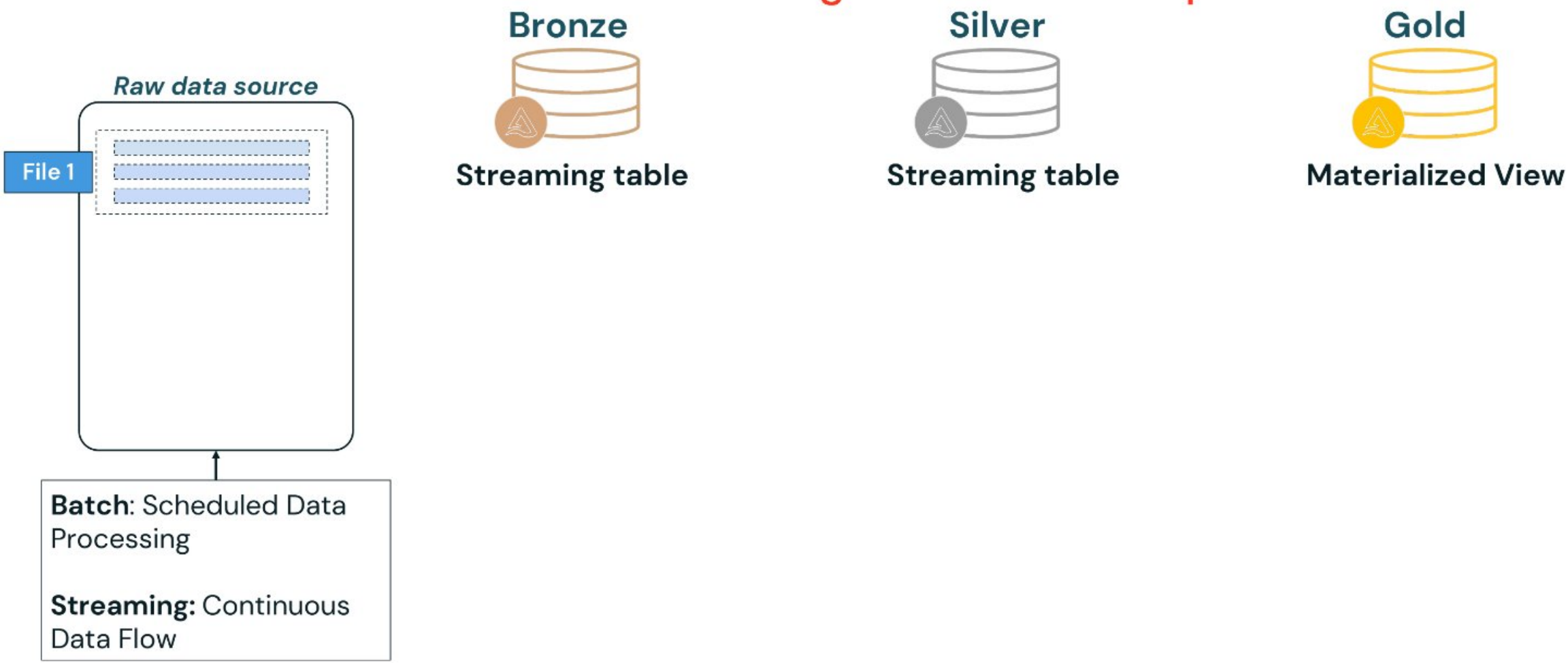
Simplifying Batch and Streaming ETL in the Medallion Architecture



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What are Spark Declarative Pipelines?

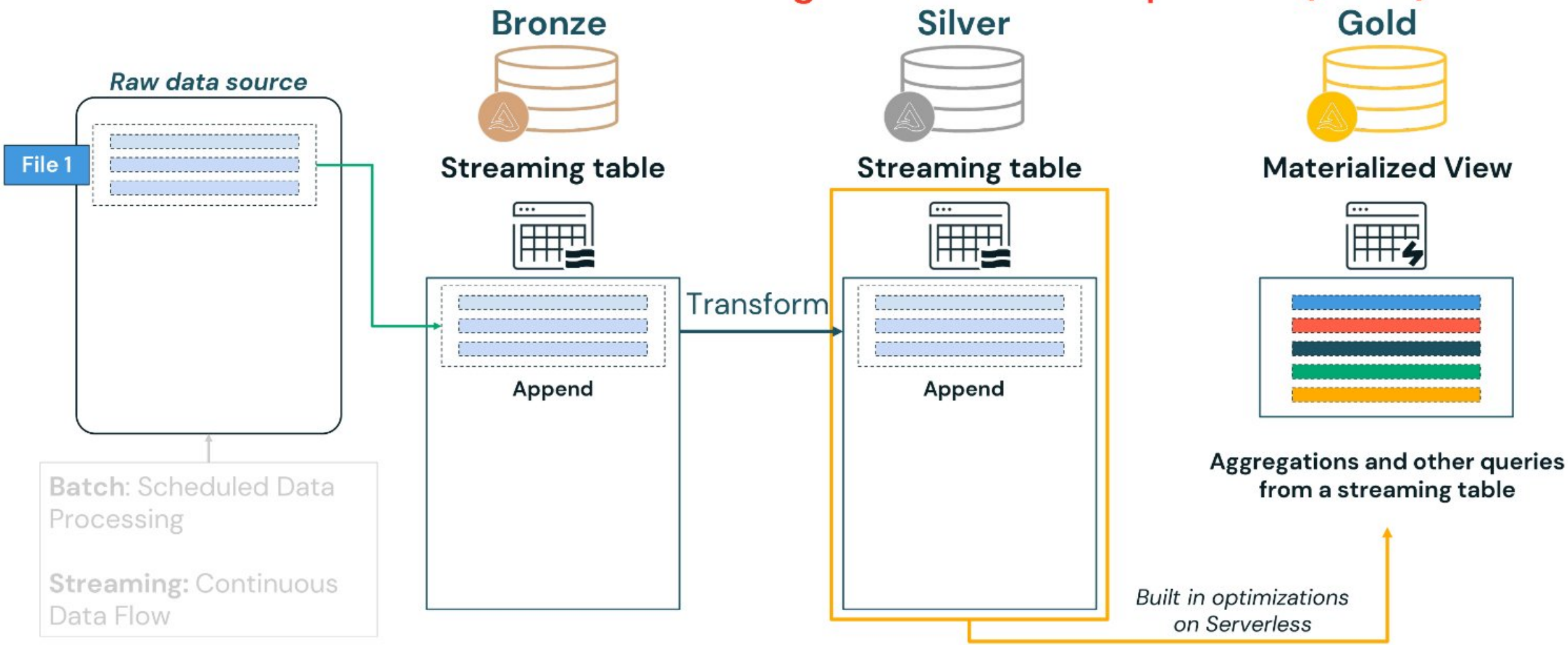
Overview of Incremental Processing in Declarative Pipelines



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What are Spark Declarative Pipelines?

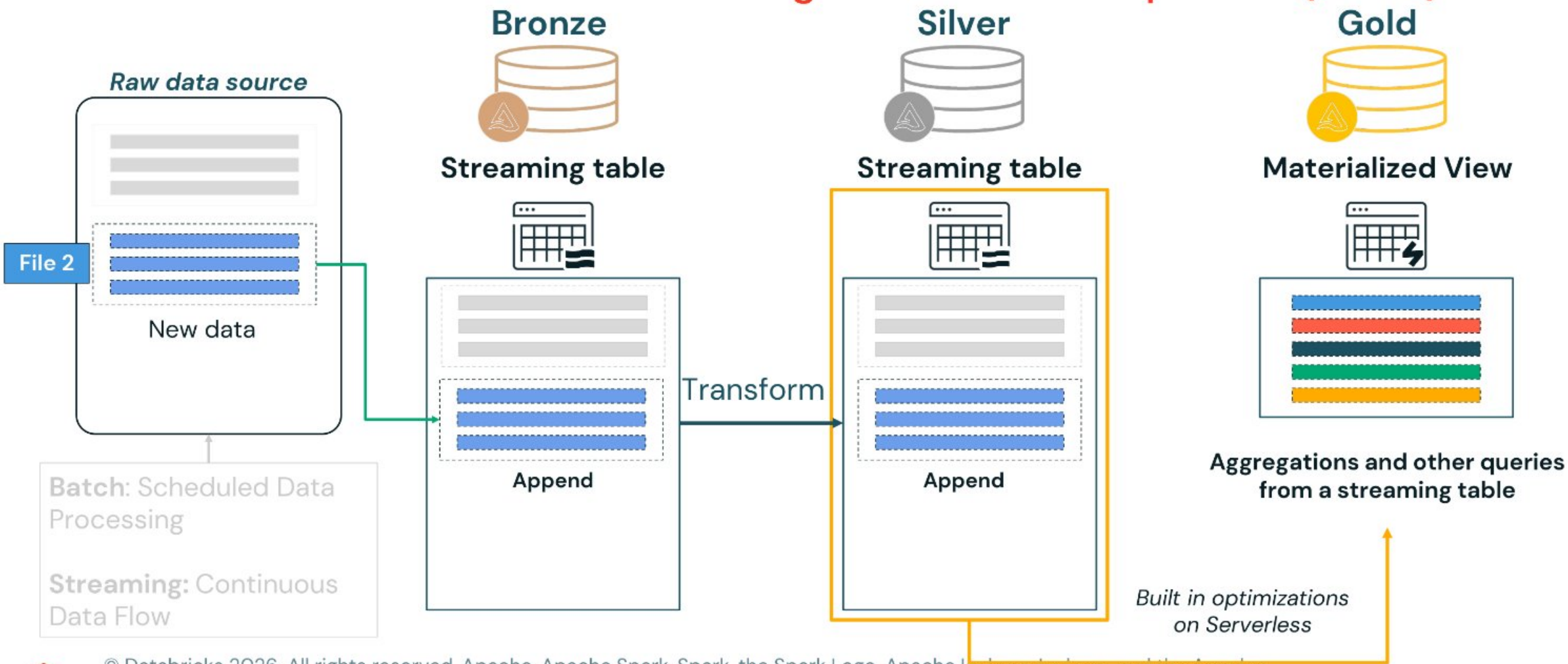
Overview of Incremental Processing in Declarative Pipelines (Run 1)



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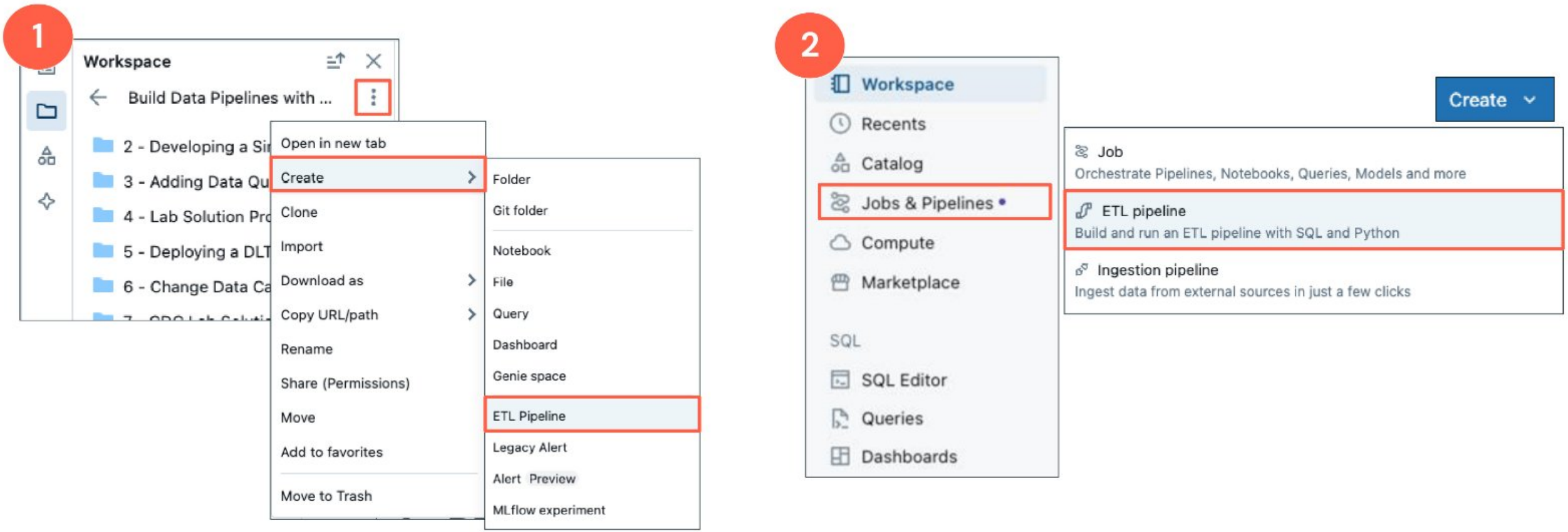
What are Spark Declarative Pipelines?

Overview of Incremental Processing in Declarative Pipelines (Run 2)



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Creating a Spark Declarative Pipeline



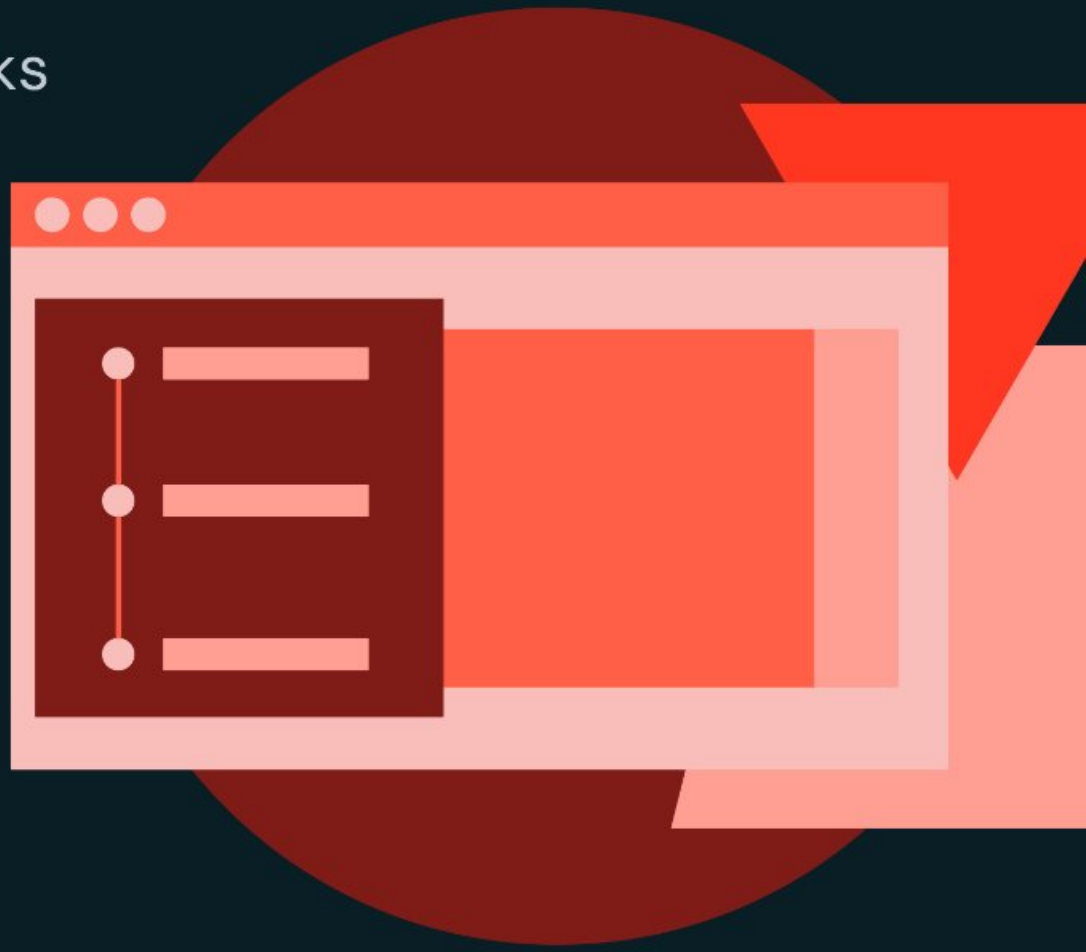
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Introduction to Data Engineering in Databricks

DEMONSTRATION

Course Setup and Creating a Pipeline



Notebook: 1 Demo - Course Setup and Creating a Pipeline

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Introduction to Data Engineering in Databricks

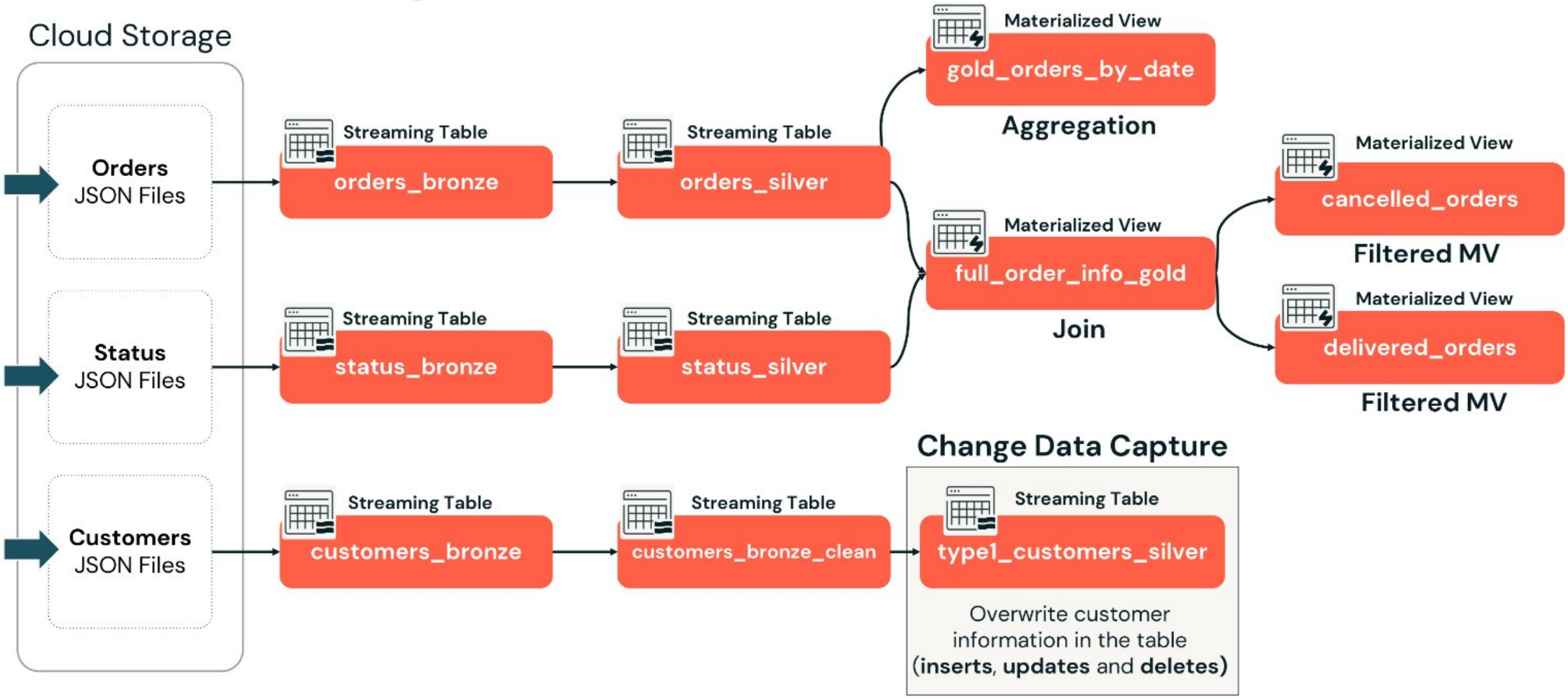
LECTURE

Course Project Overview



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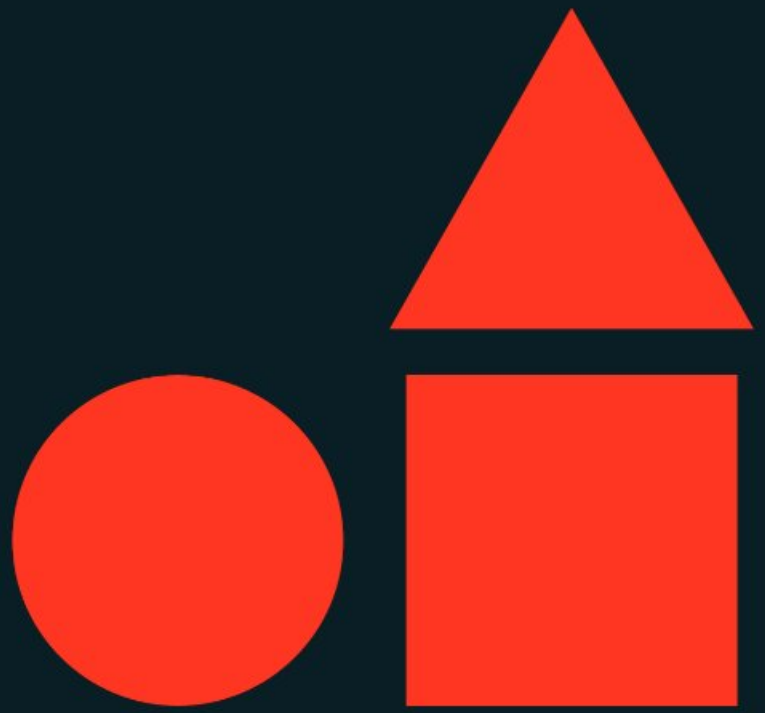
Course Project Overview



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Lakeflow Spark Declarative Pipeline Fundamentals



Build Data Pipelines with Lakeflow Spark Declarative Pipelines

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Agenda

Section Overview - Lakeflow Spark Declarative Pipeline Fundamentals
• Dataset Types Overview
• Simplified Pipeline Development
• Common Pipeline Settings
• Developing a Simple Pipeline
• Ensure Data Quality with Expectations



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Section Learning Objectives

- Understand the core concepts and components of Lakeflow Spark Declarative Pipelines pipelines, including how streaming tables, materialized views, and temporary views function and differ.
- Identify and configure pipeline settings such as compute, data assets, trigger modes, and advanced options.
- Develop a functional Declarative Pipeline using the new pipeline editor and SQL-based syntax for the pipeline.
- Incorporate data quality expectations into a Declarative Pipeline pipeline to validate and enforce data integrity.



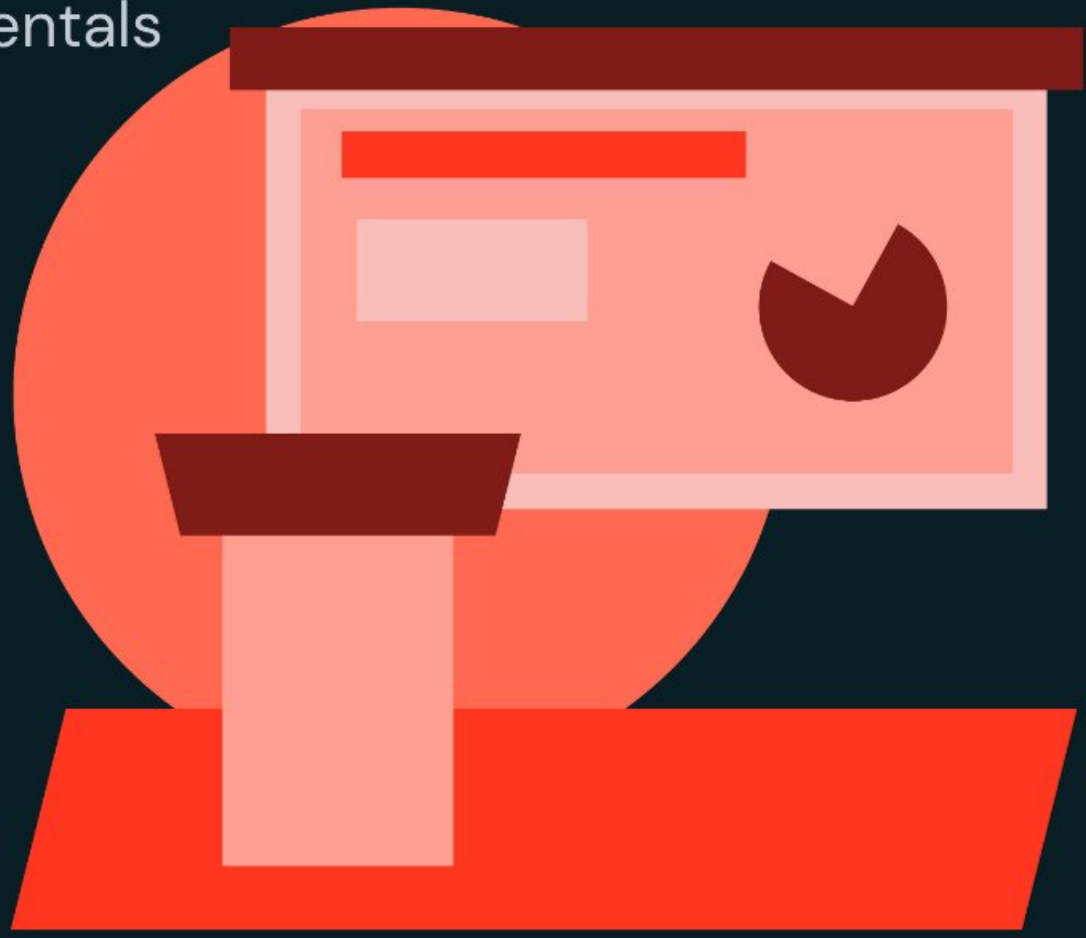
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Lakeflow Spark Declarative Pipeline Fundamentals

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Dataset Types Overview



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Dataset Types Overview

How streaming tables, materialized views, and views process data



STREAMING TABLE (ST)

A table with support for streaming or incremental data processing, **only processing new data**



MATERIALIZED VIEW (MV)



VIEWS

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Dataset Types Overview

STREAMING TABLE Overview

STREAMING TABLE

- Support for **batch** or **streaming** for **incremental** data processing (exactly **once processing** of the files)
- Each time an streaming table is refreshed, data added to the source tables is **appended** to the **streaming table**
- SQL code that creates streaming tables:
  `CREATE OR REFRESH STREAMING TABLE`
- In SQL, you must use `FROM STREAM read_files()` to **invoke Auto Loader functionality** for incremental streaming reads and checkpoints
- File names are **guaranteed to be read only once**



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Dataset Types Overview

Create a STREAMING TABLE with SQL

Create a STREAMING TABLE named **orders_bronze** from **JSON files**

```
CREATE OR REFRESH STREAMING TABLE 1_bronze_db.orders_bronze AS
SELECT
 *,
 current_timestamp() AS processing_time,
 _metadata.file_name AS source_file
FROM STREAM read_files(
   "/Volumes/dbacademy/ops/labuser/orders",
   format => 'JSON');
```

Note: The query is using the default catalog

Creates a streaming table named **orders_bronze** in the **1_bronze_db** schema

Use the SELECT clause to specify the columns to read into the streaming table from your data source

- The **STREAM** keyword indicates to use streaming read for the source files (JSON)
- The **read_files()** function reads files under a provided location and returns the data in tabular form



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Dataset Types Overview

Create a STREAMING TABLE with SQL

Create a STREAMING TABLE named **orders_silver** that reads from the **STREAMING orders_bronze** table

```
CREATE OR REFRESH STREAMING TABLE 2_silver_db.orders_silver AS
SELECT
    order_id,
  timestamp(order_timestamp) AS order_timestamp,
  customer_id,
  notifications
FROM STREAM 1_bronze_db.orders_bronze;
```

Specify your SQL transformations for the streaming data from the bronze table

Use **STREAM** keyword with the name of the STREAMING TABLE you are reading from to transform new rows

Note: The query is using the default catalog

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Dataset Types Overview

How streaming tables, materialized views, and views process data



STREAMING TABLE (ST)

A table with support for streaming or incremental data processing, **only processing new data**



MATERIALIZED VIEW (MV)

- Records are processed as required to return accurate results for the **current data state**
- Used for data processing tasks such as:
  - Transformations
  - Aggregations
  - Pre-computing slow queries
  - Frequently used computations



VIEWS

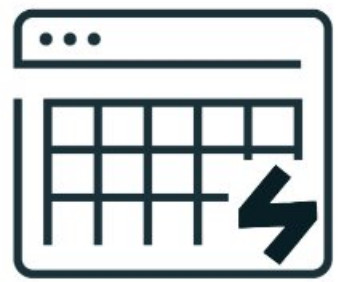
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Dataset Types Overview

MATERIALIZED VIEW Overview

MATERIALIZED VIEW

- Each time a materialized view is updated, **query results are recalculated to reflect changes in upstream datasets**
- Created and **kept up-to-date** by pipeline
- Use the `CREATE OR REFRESH MATERIALIZED VIEW` syntax
- Can be used **anywhere in your pipeline**, not just in the gold layer
- **Where applicable,** results are **incrementally refreshed for materialized views**, avoiding the need to completely rebuild the materialized view when new data arrives **(Serverless Only)**
- Incremental refresh for materialized views is a cost-based optimizer to power fast and efficient transformations for materialized views on Serverless compute



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Dataset Types Overview

Create a MATERIALIZED VIEW with SQL

Create a MATERIALIZED VIEW named **gold_orders_by_date** that summarizes the **orders_silver** table

```
CREATE OR REFRESH MATERIALIZED VIEW 3_gold_db.gold_orders_by_date AS
SELECT
 date(order_timestamp) AS order_date,
 count(*) AS total_daily_orders
FROM 2_silver_db.orders_silver
GROUP BY date(order_timestamp);
```

Create the materialized view **gold_orders_by_date** in the **3_gold_db** schema (database)

Specify the the source table **without the STREAM** keyword to utilize the source STREAMING table

NOTE: Where possible, queries will be incrementally refreshed

Note: The query is using the default catalog



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Dataset Types Overview

How streaming tables, materialized views, and views process data



STREAMING TABLE (ST)

A table with support for streaming or incremental data processing, **only processing new data**



MATERIALIZED VIEW (MV)

- Records are processed as required to return accurate results for the **current data state**
- Used for data processing tasks such:
  - Transformations
  - Aggregations
  - pre-computing slow queries
  - frequently used computations.



VIEWS

Constructs a virtual table with no physical data based on the query in your Declarative Pipelines

- TEMPORARY VIEW
- VIEW

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Dataset Types Overview

TEMPORARY VIEW Overview

TEMPORARY VIEW

- Temporary views are only **persisted across the lifetime** of the pipeline and **private to the defining pipeline**
- They are **not registered** as an object to Unity Catalog
- Use the `CREATE TEMPORARY VIEW` statement
- TEMPORARY VIEWS are useful as **intermediate queries** that **are not exposed** to end users



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Dataset Types Overview

VIEW Overview

VIEW

- Constructs a virtual table with no physical data based on the result-set of a SQL query in your pipelines
- They are **registered** as an object to Unity Catalog
- Use the `CREATE VIEW` statement

LIMITATIONS

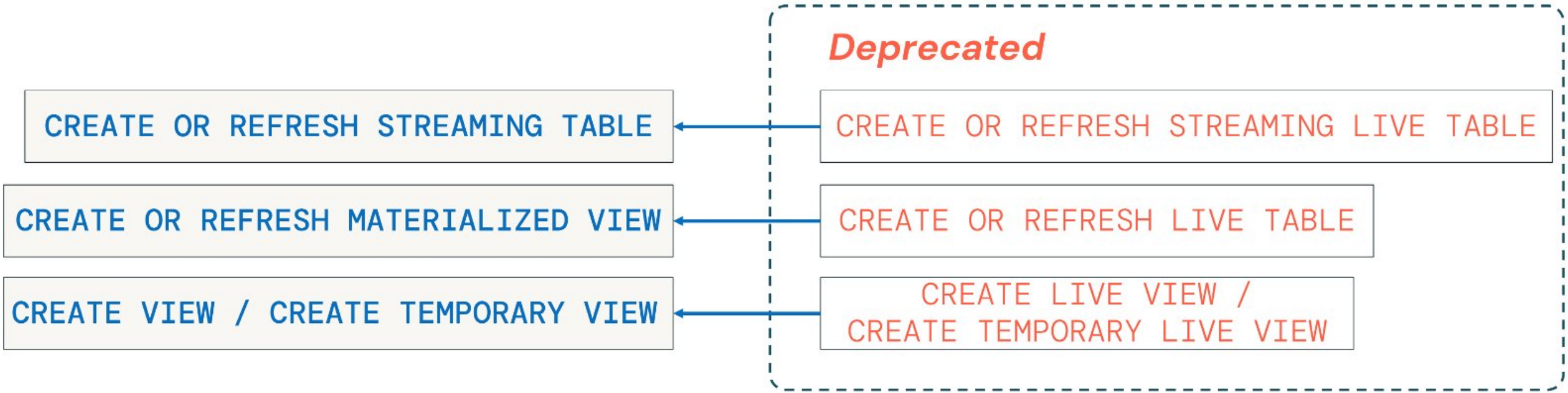
- The pipeline must be a Unity Catalog pipeline
- Views cannot have streaming queries, or be used as a streaming source for a pipeline



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Dataset Types Overview

Existing users of DLT will notice that we've evolved the names



The **semantics remain the same**, and **we'll support the old syntax** for compatibility.
Our goal is to simplify the syntax and match other systems

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Dataset Types Overview

Pipeline dependencies are parsed automatically, code order is not required

Raw JSON -> Bronze

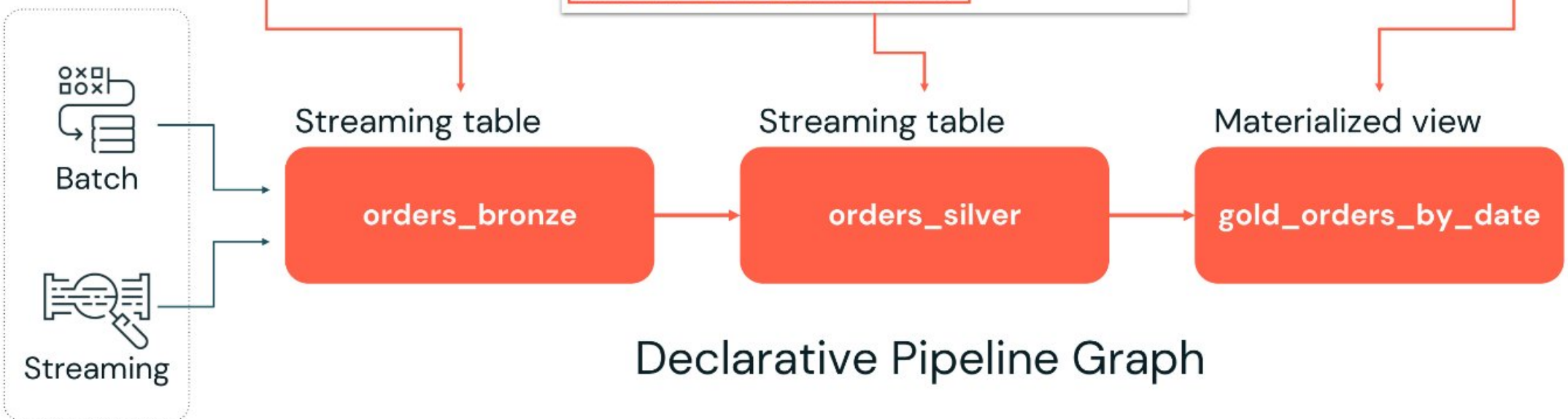
```
CREATE OR REFRESH STREAMING TABLE
1_bronze_db.orders_bronze AS
SELECT
 *,
 current_timestamp() AS processing_time,
 _metadata.file_name AS source_file
FROM STREAM read_files(
   "/Volumes/dbacademy/ops/labuser/orders",
   format => 'JSON');
```

Bronze -> Silver

```
CREATE OR REFRESH STREAMING TABLE 2_silver_db.orders_silver
AS
SELECT
  order_id,
  timestamp(order_timestamp) AS order_timestamp,
  customer_id,
  notifications
FROM STREAM 1_bronze_db.orders_bronze;
```

Silver -> Materialized View

```
CREATE OR REFRESH MATERIALIZED VIEW 3_gold_db.gold_orders_by_date
AS
SELECT
 date(order_timestamp) AS order_date,
 count(*) AS total_daily_orders
FROM 2_silver_db.orders_silver
GROUP BY date(order_timestamp);
```



Declarative Pipeline Graph

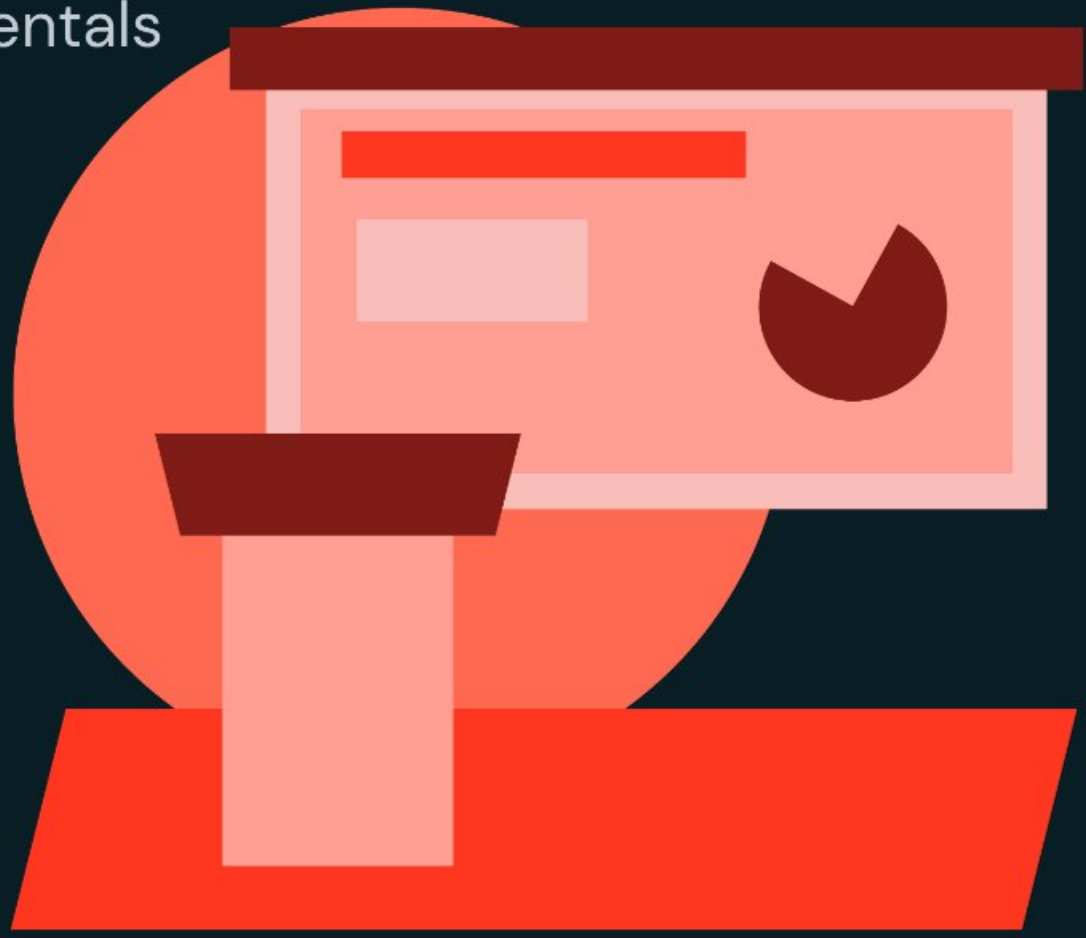
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Lakeflow Spark Declarative Pipeline Fundamentals

LECTURE

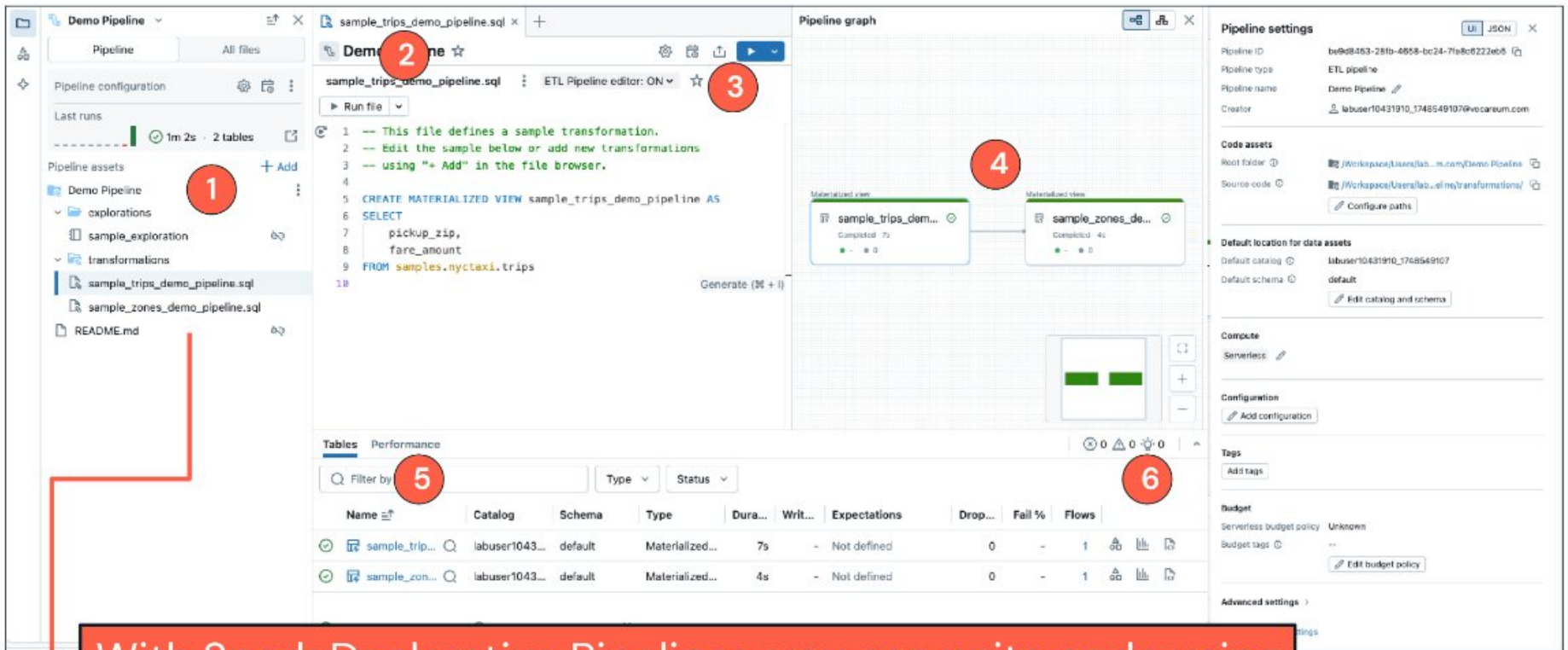
Simplified Pipeline Development



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Simplified Pipeline Development

Multi-file editor in Lakeflow Spark Declarative Pipelines



With Spark Declarative Pipelines, you can write code using **.sql or .py files (recommended)**, or use notebooks.

What's New

1. Pipeline asset browser
2. Multi-file code editor with features for step-by-step development
3. Pipeline-specific toolbars
4. Interactive DAG
5. Data previews
6. Execution insights panels
   - Easier debugging
   - Faster validation with dry run

As of Data & AI Summit 2025

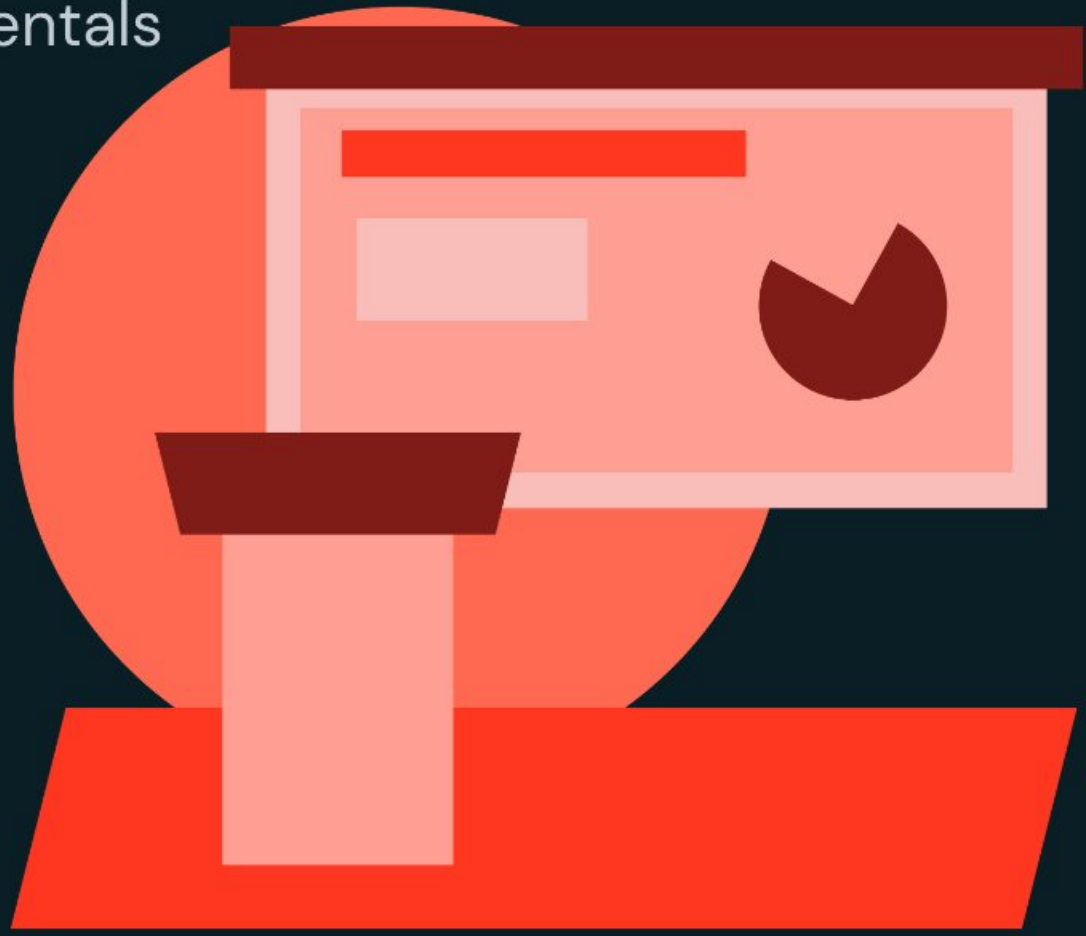
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LECTURE

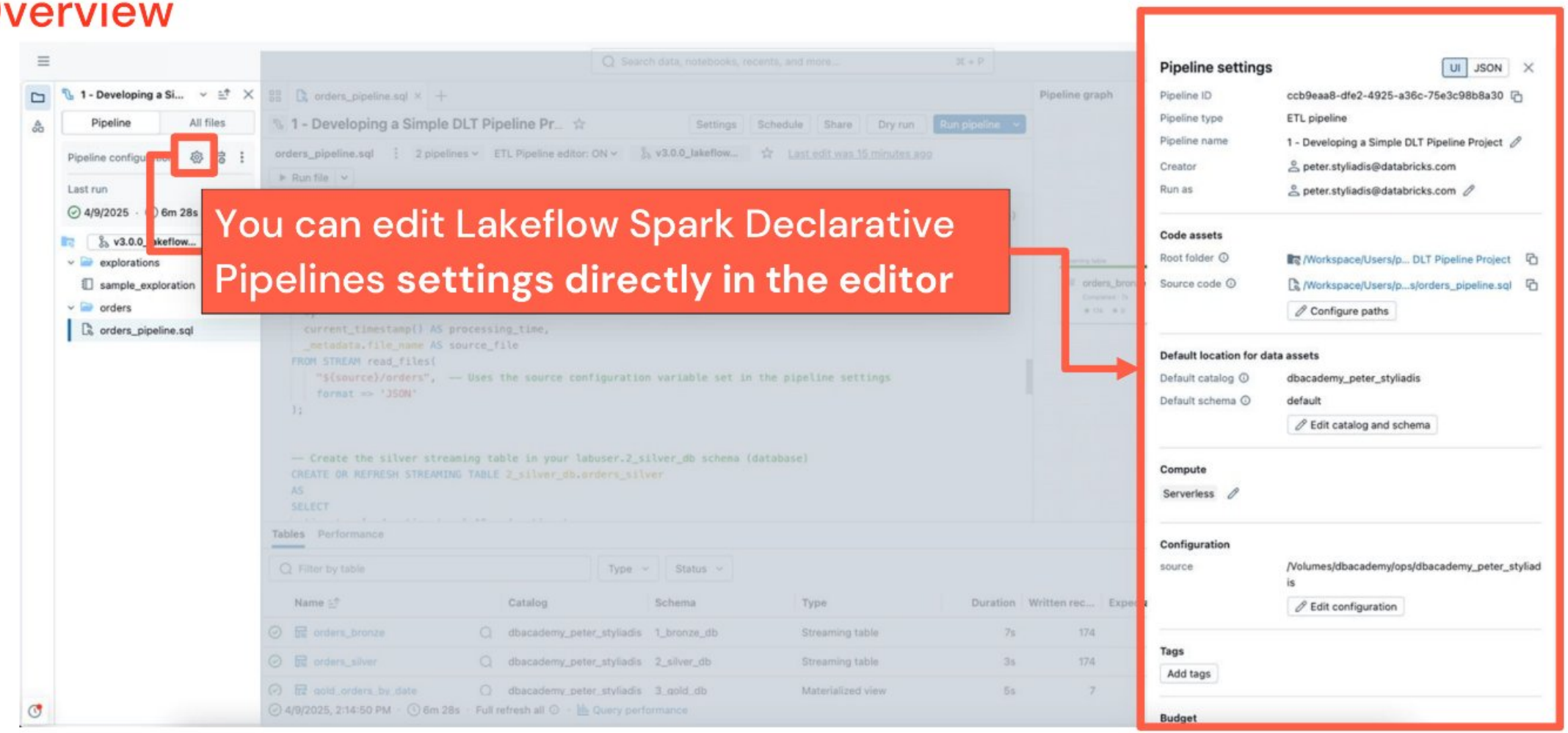
Common Pipeline Settings



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Common Pipeline Settings

Overview



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Common Pipeline Settings



Compute



Code Assets



Configuration
(parameters)

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Common Pipeline Settings



Compute



Code Assets



Configuration
(parameters)

SERVERLESS

- Databricks **recommends** developing new pipelines using serverless
- **Optimizes** cost while **maintaining** performance
- **Enables** you to focus on code rather than managing infrastructure
- Optional **Serverless Performance optimized** setting for time-sensitive Lakeflow Jobs

- Serverless compute includes **Incremental refresh for materialized views**
- A cost-based optimizer to power fast and efficient transformations for materialized views on Serverless compute

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Common Pipeline Settings



Compute



Code Assets



Configuration
(parameters)

SERVERLESS

- Databricks **recommends** developing new pipelines using serverless
- **Optimizes** cost while **maintaining** performance
- **Enables** you to focus on code rather than managing infrastructure
- Optional **Serverless Performance optimized** setting for time-sensitive Lakeflow Jobs

CLASSIC (fixed size)

- Users need **permission** to create compute for Declarative Pipelines
- Workspace admins **can configure cluster policies** to provide users with access to compute resources for Declarative Pipelines

AUTOSCALING

- Enhanced autoscaling **is enabled by default** for all new pipelines.
- **Optimizes cluster utilization** by automatically allocating cluster resources based on workload volume



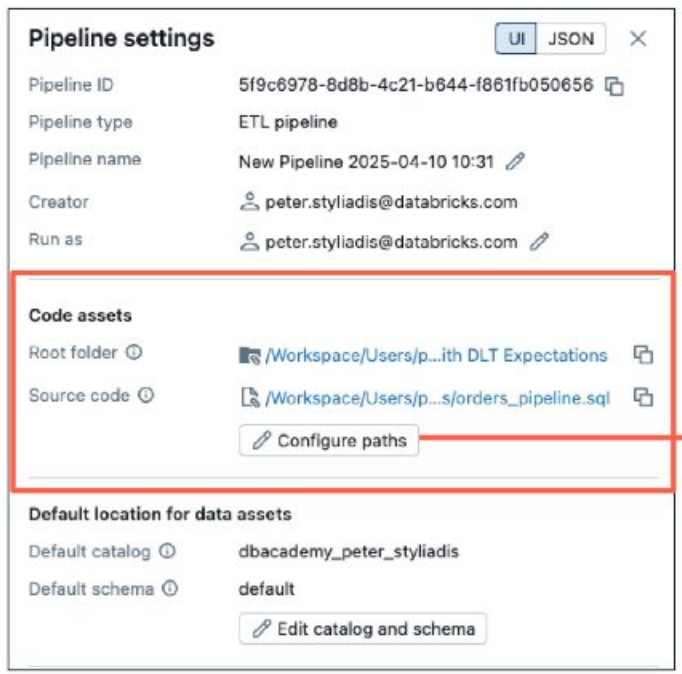
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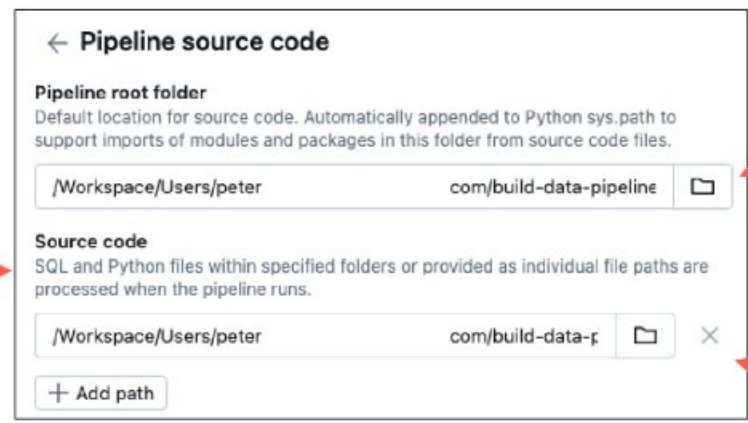
Common Pipeline Settings

Compute

Code Assets

Configuration
(parameters)





The **Pipeline root folder** automatically includes all relevant files within the folder for the pipeline project (can be a Git folder)

The **source code** section references include subfolders (or files) for the pipeline **(.py, .sql, notebooks)**

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Common Pipeline Settings

Compute

Code Assets

Configuration
(parameters)

A pipeline's configuration is **a map of key value pairs** that can be used to parameterize your code:

- Improve code readability/maintainability
- Reuse common parameters in multiple pipelines files




```
CREATE STREAMING TABLE bronze
SELECT *
FROM STREAM read_files(
  "${source}/orders",
  format => 'JSON');
```

In SQL reference the key's value using the **${key-name}** syntax

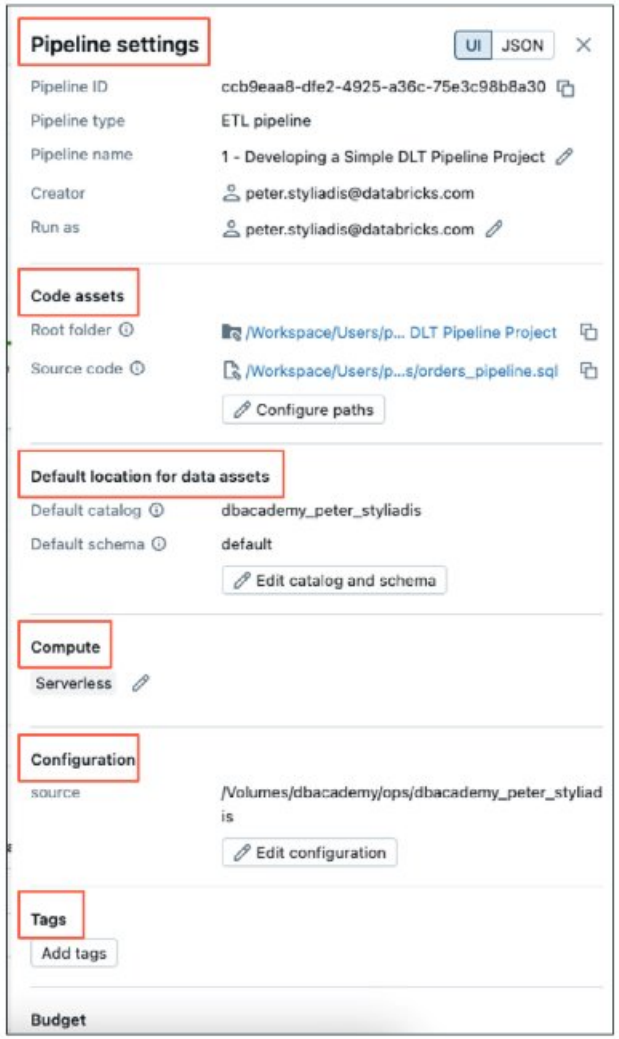
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Common Pipeline Settings

Common Settings

There are **various other settings** we will explore throughout the course, including:

- Pipeline settings
- Code assets
- Default location for data assets
- Compute
- Configuration
- Environment
- Tags
- Budget
- Advanced settings



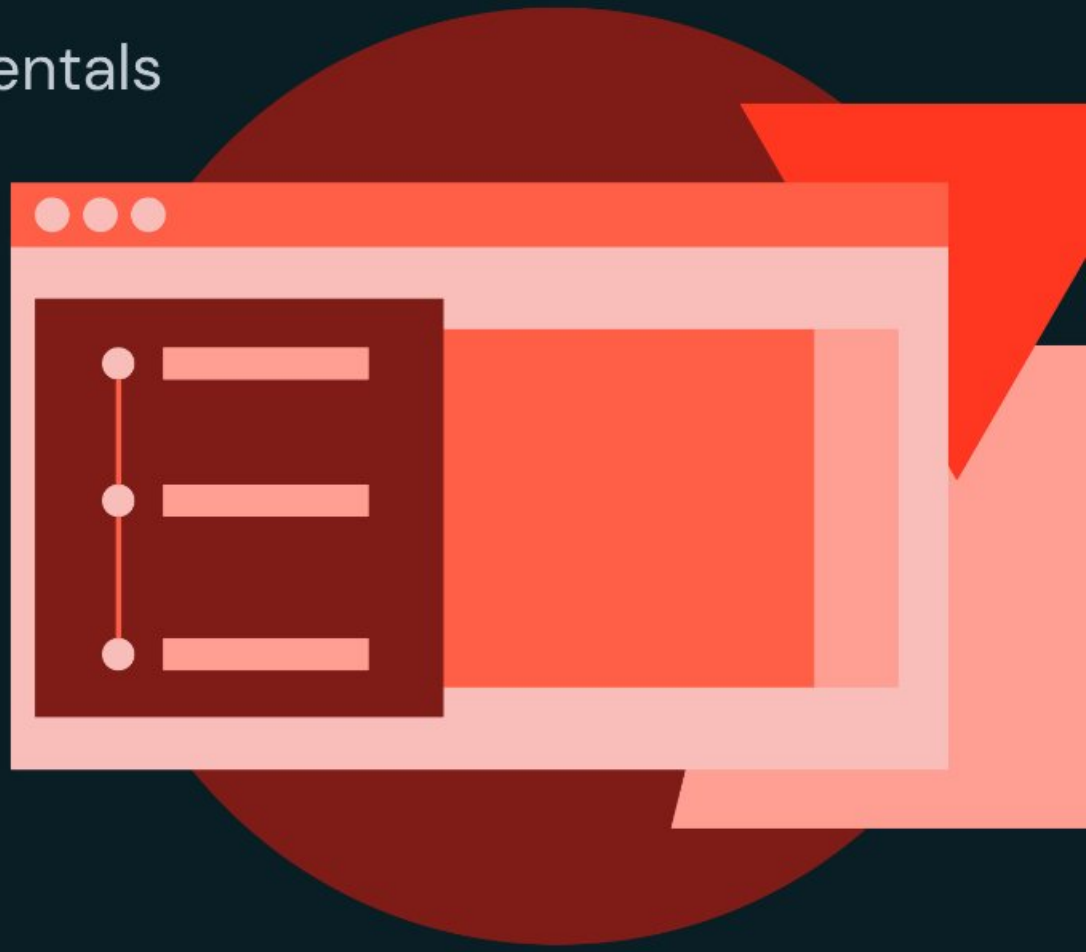
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Lakeflow Spark Declarative Pipeline Fundamentals

DEMONSTRATION

Developing a Simple Pipeline



Notebook: 2 Demo - Developing a Simple Pipeline

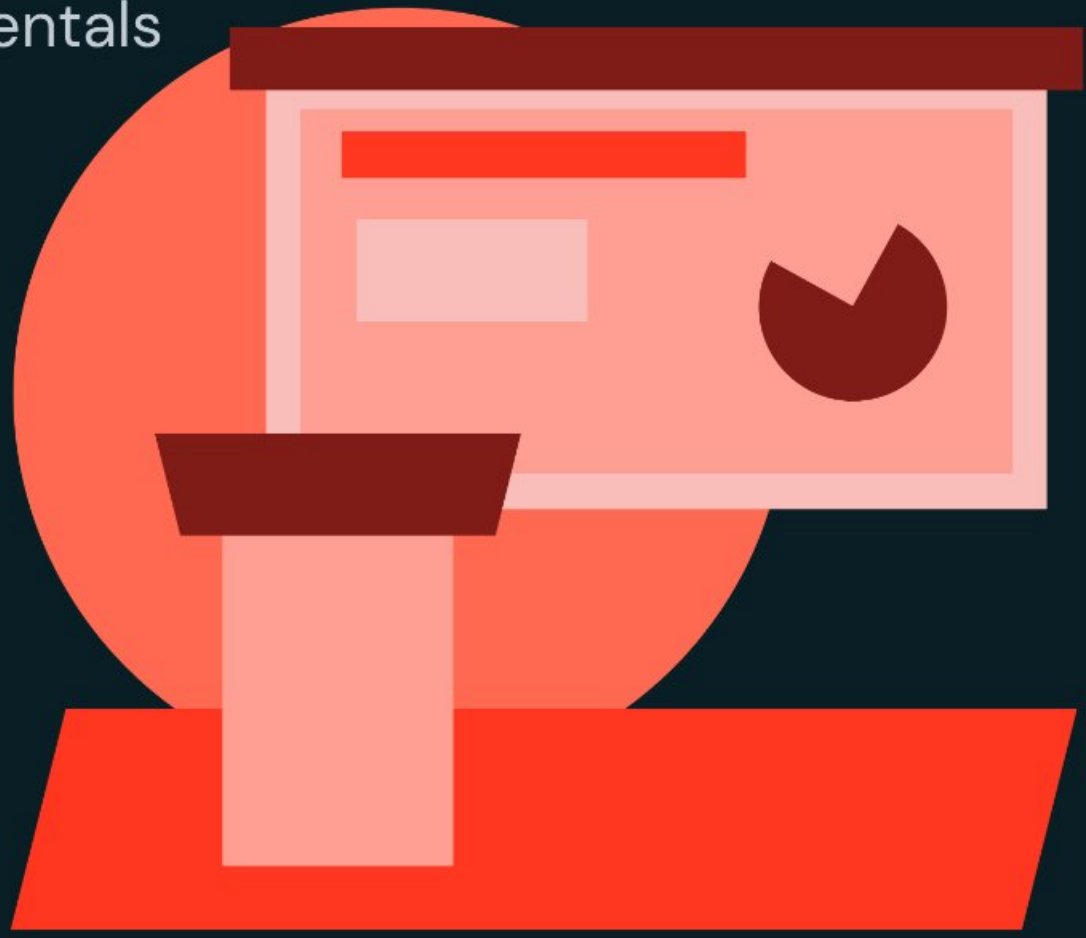
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Lakeflow Spark Declarative Pipeline Fundamentals

LECTURE

Ensure Data Quality with Expectations



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Ensure Data Quality with Expectations

Creating Expectations

With Lakeflow Spark Declarative Pipelines you can apply **data quality rules** to validate rows during ETL processes to gain insights into data integrity using the following

```
CONSTRAINT constraint-name EXPECT (column condition) [ON VIOLATION action]
```

Action	SQL Syntax Example	Result
WARN (default)	`CONSTRAINT valid_notification EXPECT (notifications IN ('Y','N'))`	• Invalid rows **are still written** to the target • Logs include **counts of valid vs. invalid records** and other metrics
DROP	`CONSTRAINT valid_date EXPECT (order_timestamp > "2021-01-01")` `ON VIOLATION DROP ROW`	• Invalid rows **dropped** from table • The count is logged alongside other metrics
FAIL	`CONSTRAINT valid_id EXPECT (customer_id IS NOT NULL)` `ON VIOLATION FAIL UPDATE`	• Causes a **failure of a single flow** and does not cause other flows in your pipeline to fail. • **Manual** intervention is required



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Ensure Data Quality with Expectations

Adding Data Quality Expectations Example

```
CREATE OR REFRESH STREAMING TABLE 2_silver_db.orders_silver
 (
   CONSTRAINT valid_notifications EXPECT (notifications IN ('Y','N')),
   CONSTRAINT valid_date EXPECT (order_timestamp > "2021-01-01") ON VIOLATION FAIL UPDATE,
   CONSTRAINT valid_id EXPECT (customer_id IS NOT NULL) ON VIOLATION DROP ROW
 )
AS
SELECT
      Order_id,
   timestamp(order_timestamp) AS order_timestamp,
   Customer_id,
   notifications
FROM STREAM 1_bronze_db.orders_bronze;
```

WARN in metrics

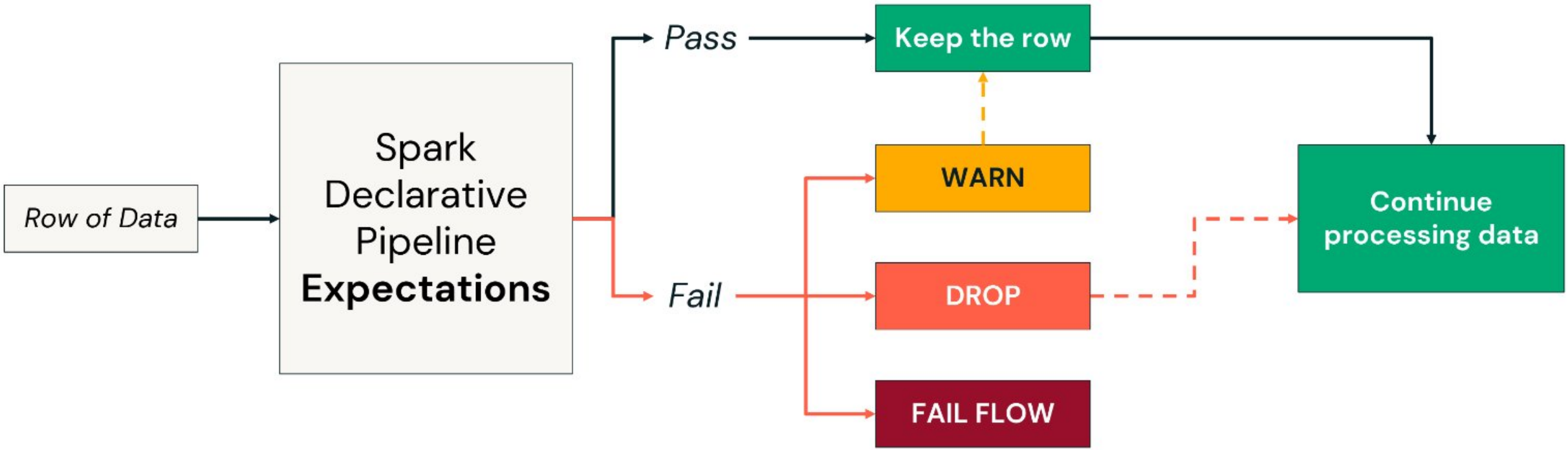
DROP the row

FAIL THE PIPELINE

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Ensure Data Quality with Expectations

Actions Overview



As of 2025Q2 materialized views that use **expectations** are **always fully refreshed**

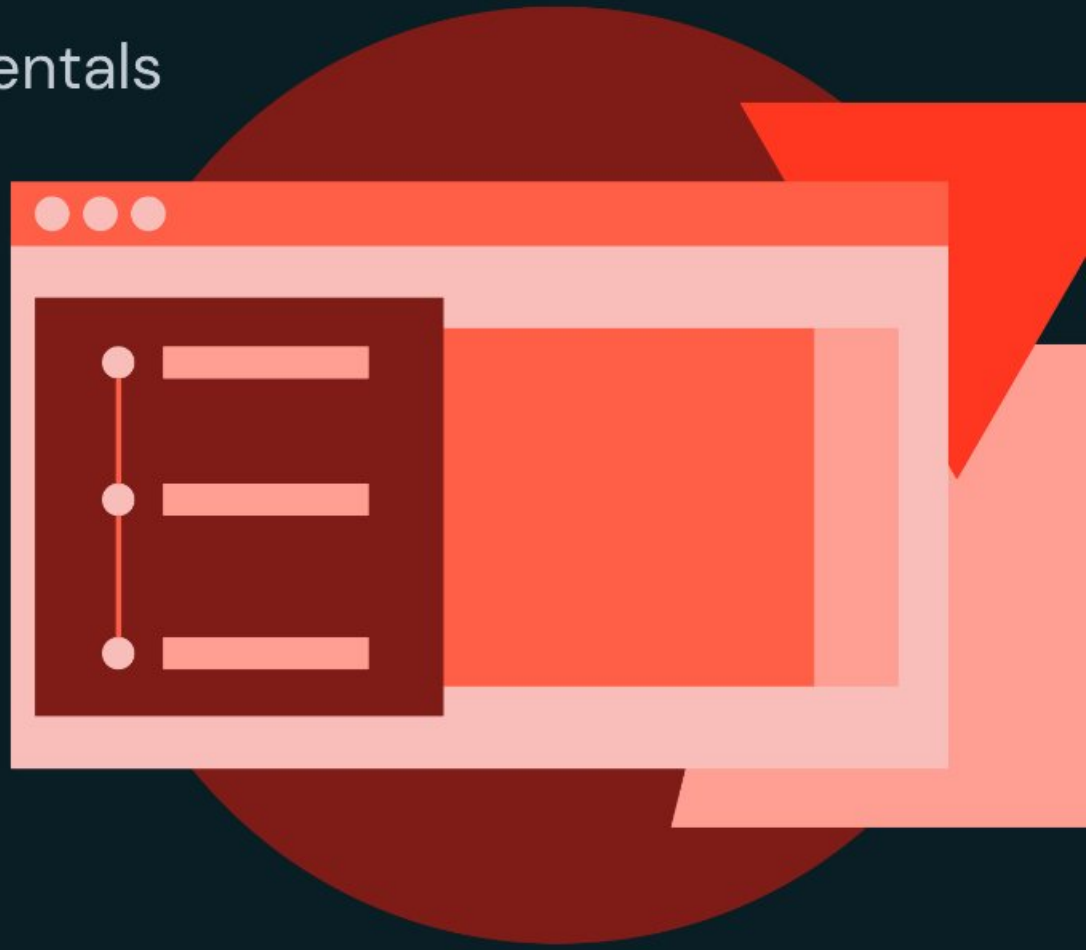
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Lakeflow Spark Declarative Pipeline Fundamentals

DEMONSTRATION

Adding Data Quality Expectations



Notebook: 3 Demo - Adding Data Quality Expectations

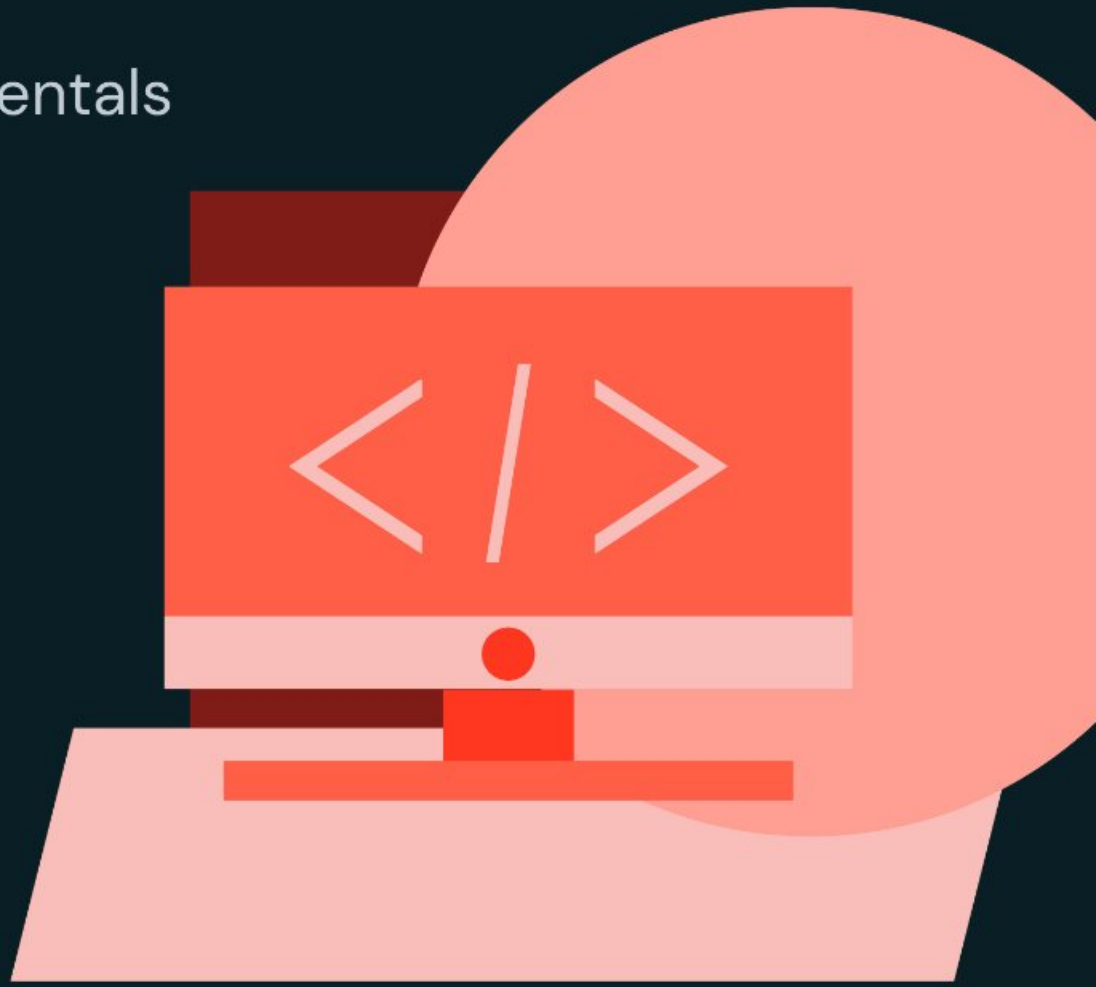
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Lakeflow Spark Declarative Pipeline Fundamentals

LAB EXERCISE

Create a Pipeline

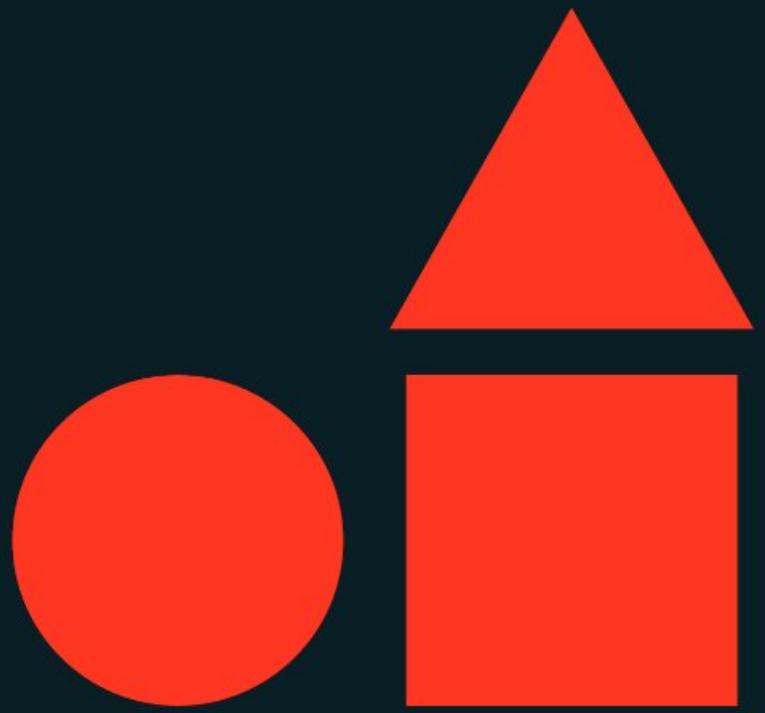


Notebook: 4 Lab - Create a Pipeline

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Building Lakeflow Spark Declarative Pipelines



Build Data Pipelines with Lakeflow Spark Declarative Pipelines

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Agenda

Section Overview - Building Pipelines
• Streaming Joins Overview
• Deploying a Pipeline to Production
• Change Data Capture (CDC) Overview
• Change Data Capture with AUTO CDC INTO
• Additional Features Overview



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Section Learning Objectives

- Deploy a Lakeflow Spark Declarative Pipeline in production by modifying configuration options like mode, schedule, email notifications and more.
- Analyze event logs and pipeline metrics to examine the entirety of a pipeline.
- Design and implement a Change Data Capture (CDC) Declarative Pipeline using AUTO CDC INTO to handle slowly changing dimensions (SCD).



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Building Lakeflow Spark Declarative Pipelines

LECTURE

Streaming Joins Overview



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Streaming Joins Overview

Stream-Snapshot Join Overview

Streaming Table to **Static Table**



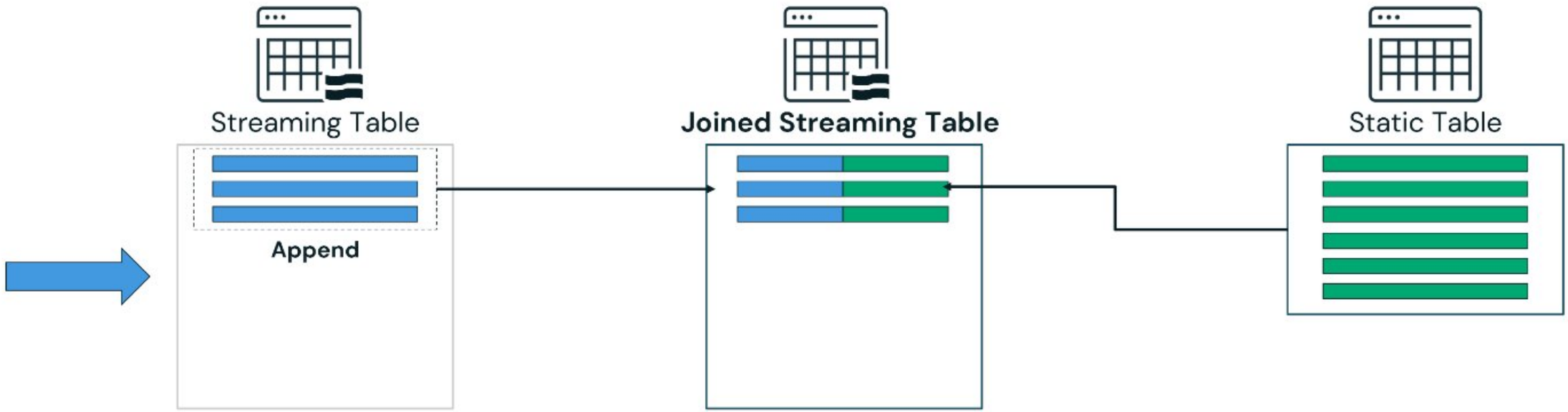
The goal is to **incrementally** join new data with a **static table**

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Streaming Joins Overview

Stream-Snapshot Join Overview

Streaming Table to **Static Table**



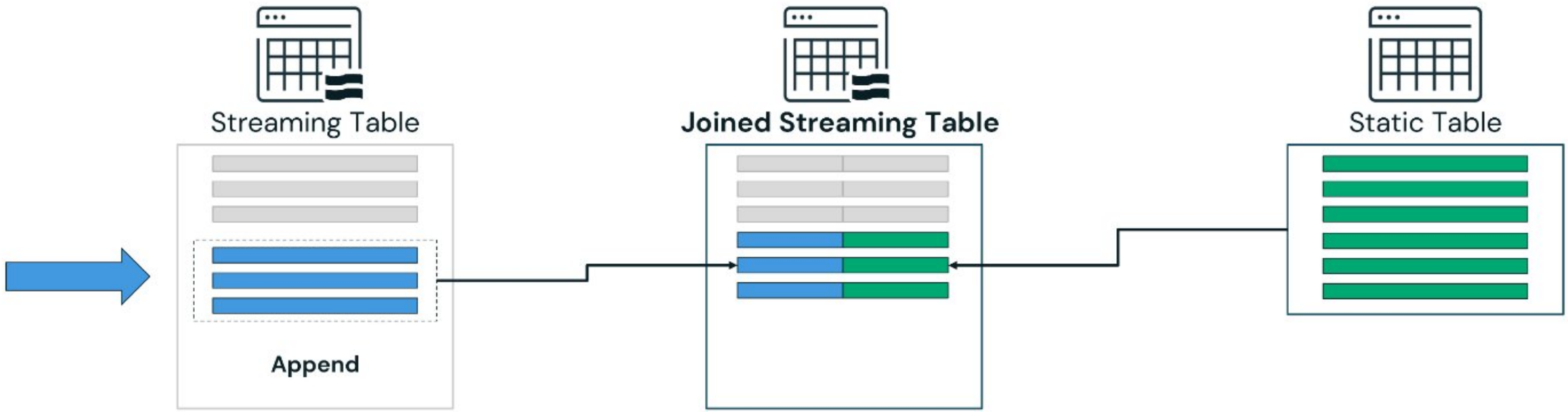
The **new data** in the streaming table JOINS using all of the **data in the lookup table**

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Streaming Joins Overview

Stream-Snapshot Join Overview

Streaming Table to **Static Table**



Only the new data in the streaming table JOINS using all of the **data in the lookup table**

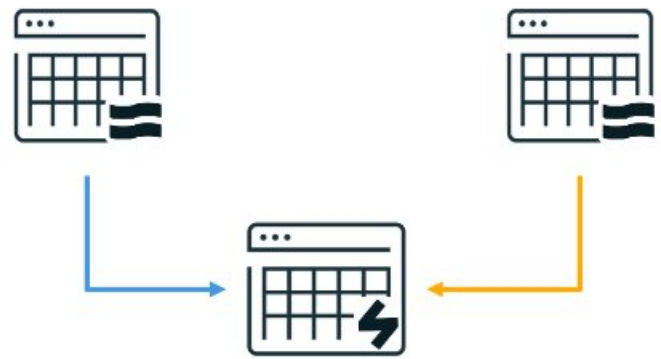
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Streaming Joins Overview

Joining Streaming Tables with a Materialized View

Streaming Table to Static Table

Streaming Table to **Streaming Table** with a **Materialized View**



The goal is to take **all rows from two streaming tables** and join them together each time the pipeline is run

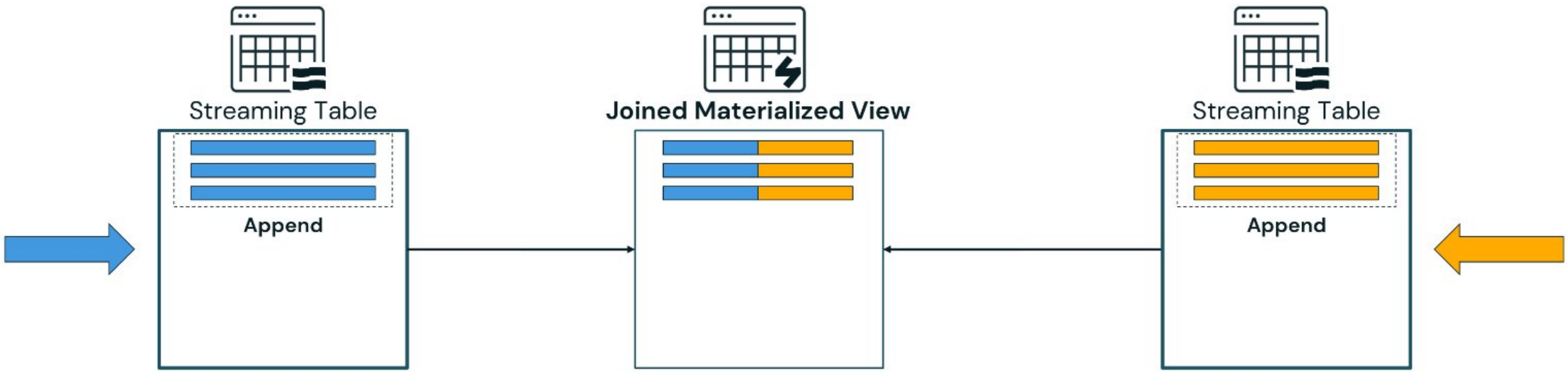
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Streaming Joins Overview

Joining Streaming Tables with a Materialized View

Streaming Table to Static Table

Streaming Table to **Streaming Table** with a **Materialized View**



The materialized view **efficiently computes all the data** in the streaming tables and joins the rows.

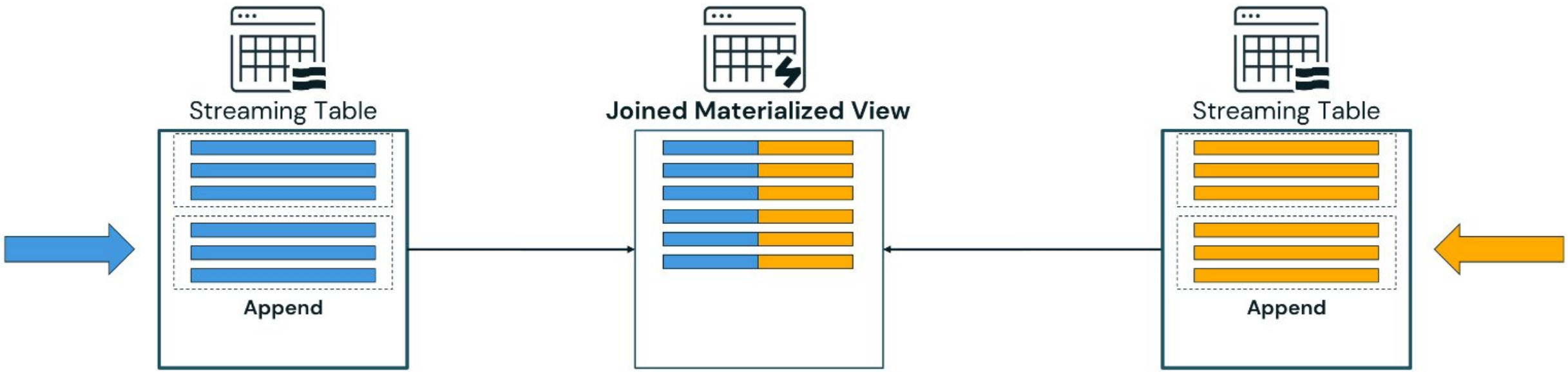
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Streaming Joins Overview

Joining Streaming Tables with a Materialized View

Streaming Table to Static Table

Streaming Table to **Streaming Table** with a **Materialized View**



As new data is added to the streaming tables, the materialized view will again **efficiently compute all the data** in the streaming tables and joins the rows

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Streaming Joins Overview

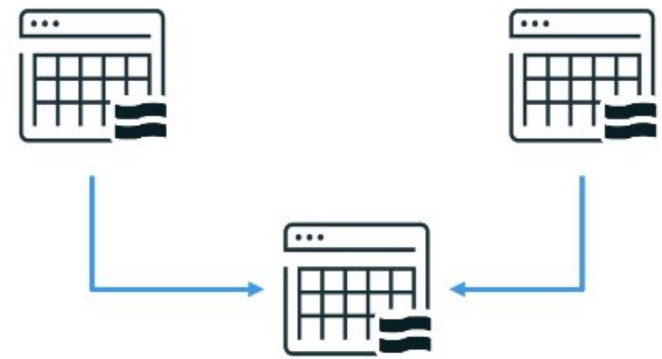
Stream-Stream Joins

Outside the scope of the course

Streaming Table to Static Table

Streaming Table to Streaming Table with a Materialized View

Incremental Stream to Stream Joins



The goal is to **incrementally** join new data from two tables, **past data is not used**

Databricks Streaming and DLT (Advanced) Course

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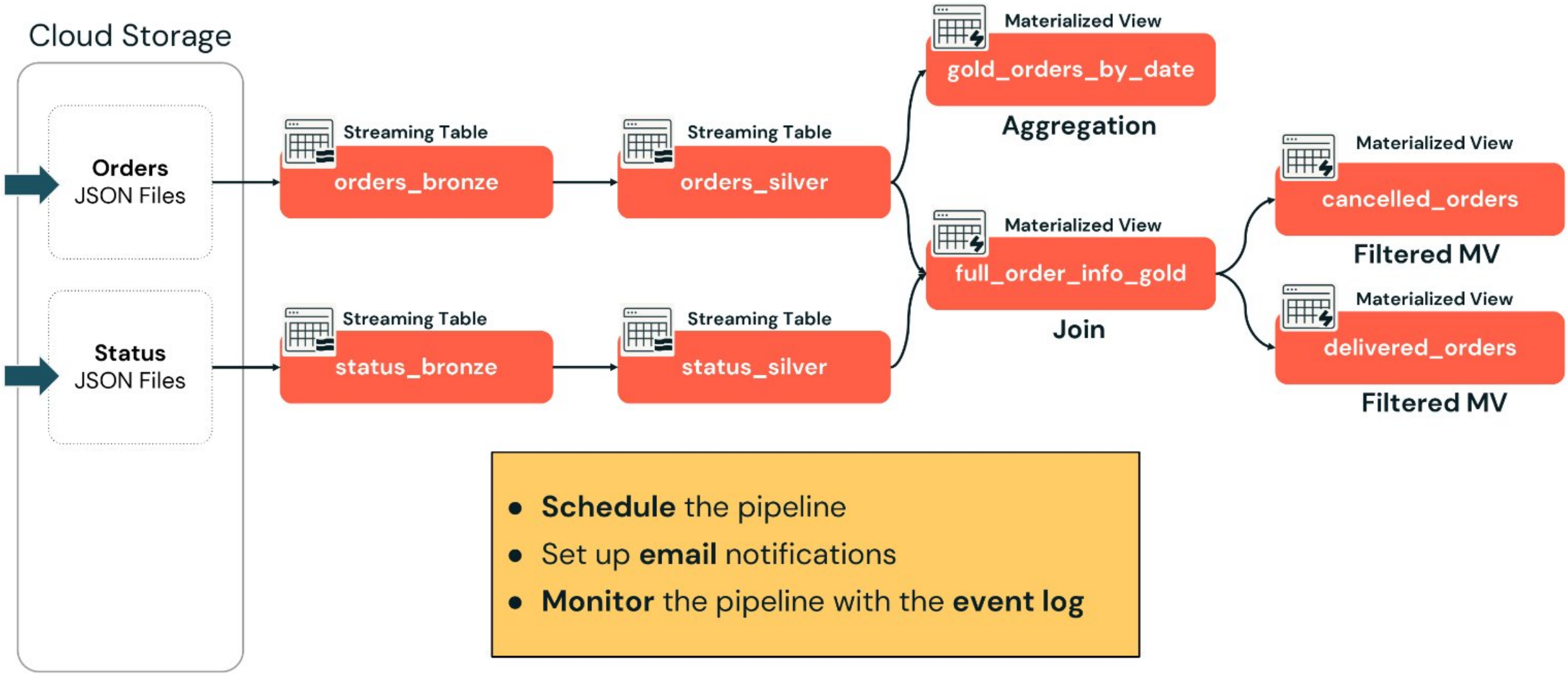
LECTURE

Deploying a Pipeline to Production



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Deploying a Pipeline to Production



- **Schedule** the pipeline
- Set up **email** notifications
- **Monitor** the pipeline with the **event log**

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Deploying a Pipeline to Production

Common Production Tasks



Scheduling



Email Notifications



Monitor pipelines

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Deploying a Pipeline to Production

Common Production Tasks



Scheduling



Email Notifications



Monitor pipelines

Triggered

- Can be triggered **manually** or run on a **schedule**
- **Refreshes** selected tables using data available at the start, then stops once the update is complete

Continuous

- **Continuously** processes new data to keep streaming tables and materialized views up to date in **near real time**
- Monitors dependencies and **updates only** when source data changes

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Deploying a Pipeline to Production

Common Production Tasks



Scheduling

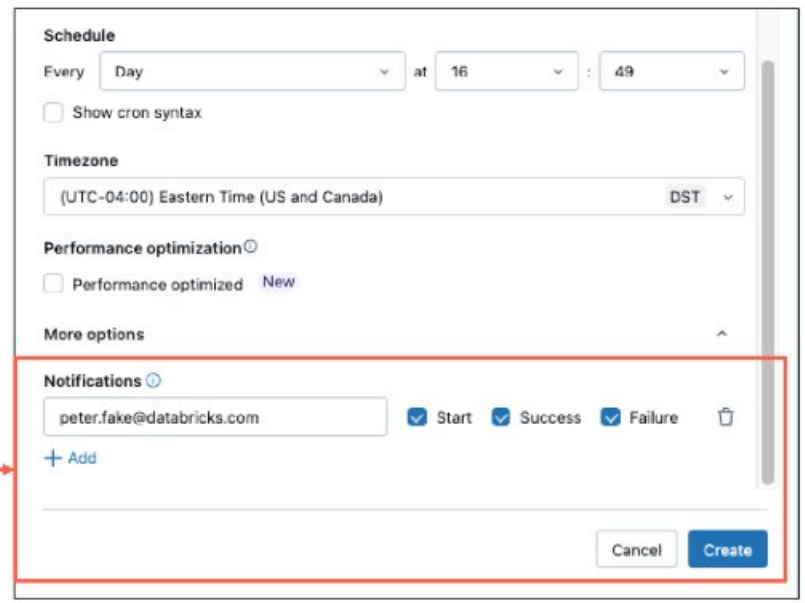


Email Notifications



Monitor pipelines

You can configure **email notifications** when **scheduling** the pipeline



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Deploying a Pipeline to Production

Common Production Tasks



Scheduling



Email Notifications



Monitor pipelines

- The **pipeline event log** captures all key information about the pipeline:
  - Audit logs
  - Data Quality Checks
  - Pipeline Progress
  - Data Lineage



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Deploying a Pipeline to Production

Querying the Declarative Pipeline Event Log

- **Publish the event log** as a **Delta table** using the **advanced settings**
  - Specify the table **location** (catalog, schema)
  - Define the **table** name

Query the pipeline event log table

```
SELECT * FROM <catalog>.<schema>.<event_log_table_name>
```



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Deploying a Pipeline to Production

Querying the Declarative Pipeline Event Log

- **Publish the event log** as a **Delta table** using the **advanced settings**
  - Specify the table **location** (catalog, schema)
  - Define the **table** name

- **(DEFAULT)** The Event Log is written as a **hidden Delta table**
  - Located in the pipeline **default catalog** and **schema**
  - View the query the event log documentation to access the hidden event log

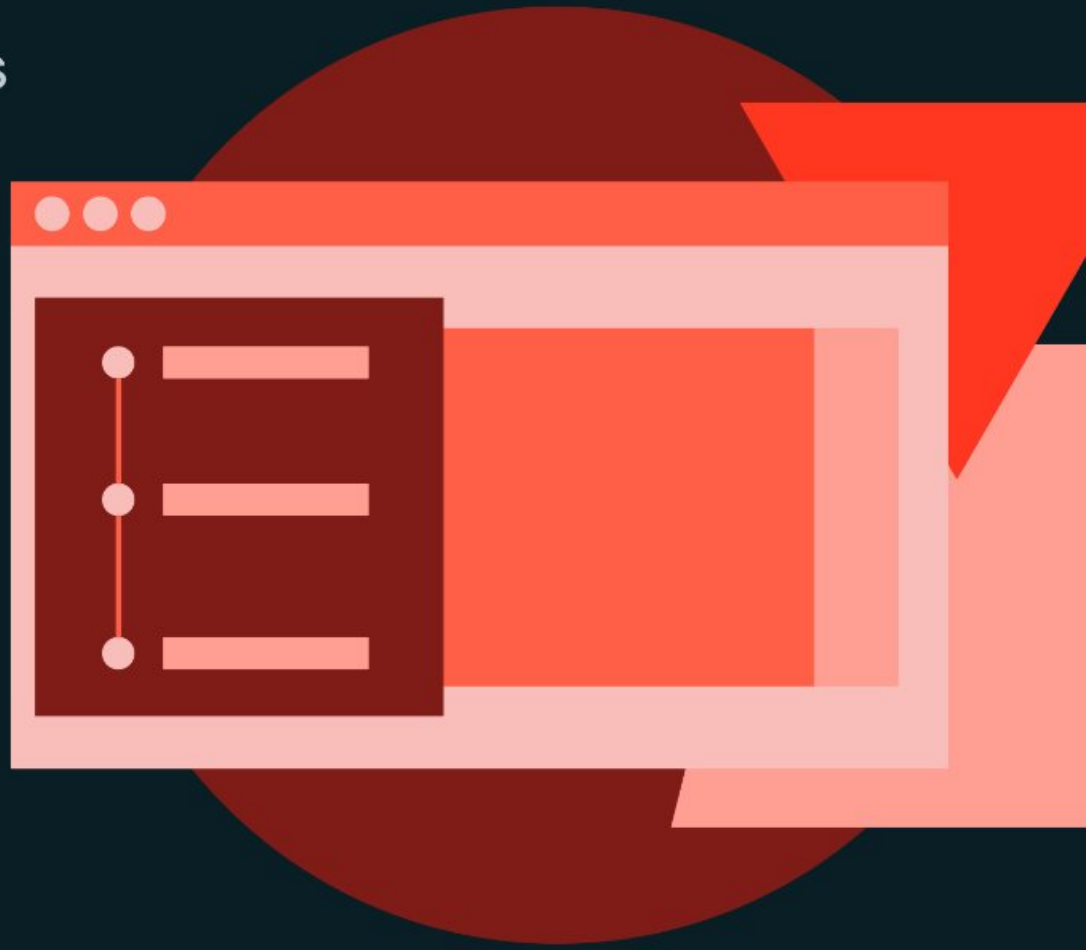
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Building Lakeflow Spark Declarative Pipelines

DEMONSTRATION

Deploying Your Pipeline to Production



Notebook: 5 Demo - Deploying a Pipeline to Production

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Building Lakeflow Spark Declarative Pipelines

LECTURE

Change Data Capture (CDC) Overview

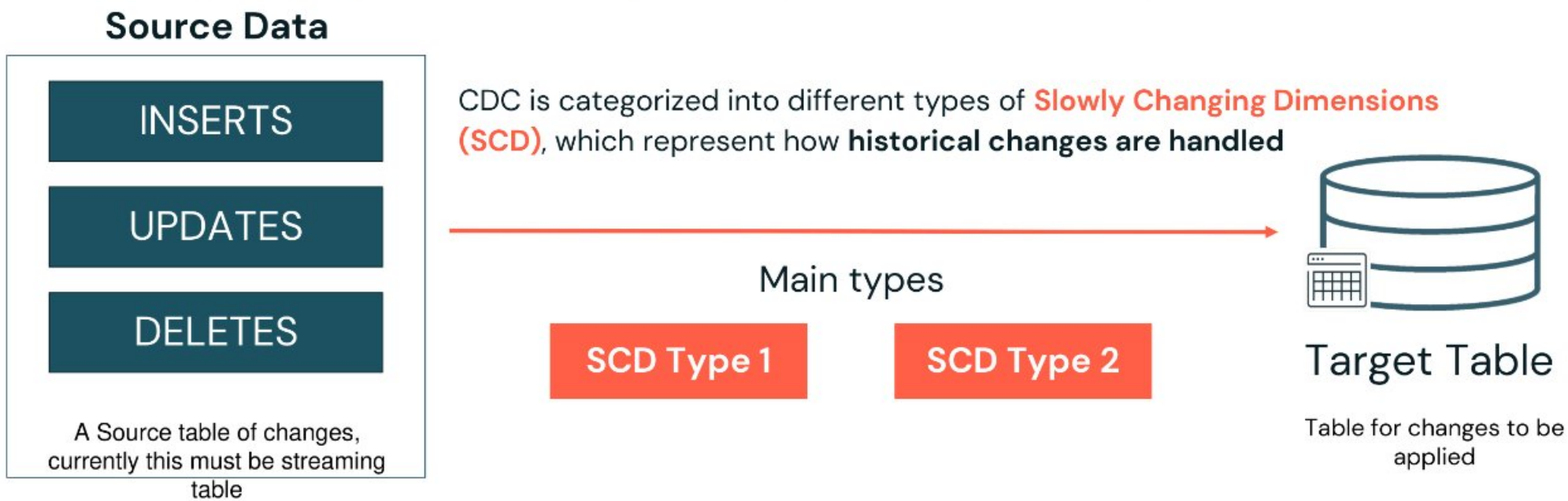


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Change Data Capture (CDC) Overview

What is Change Data Capture?

- Change Data Capture (CDC) is a technique **used to track and capture changes in a data source** (such as a database, lakehouse or data warehouse).
- Then **applying those changes** into the Lakehouse or data warehouse



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Change Data Capture (CDC) Overview

Slowing Changing Dimensions (SCD) Type 1 (1 of 2)

SCD Type 1 (Overwrite Target with Latest Values)

- When a record updates, the **previous record is simply overwritten** by its **key** with the new value.
- When a record is deleted by its **key**, the **record is removed**
- There is **no tracking of old keys (rows),** only the **current data** is retained.

customers (*target table with all customers*)

CustomerID	Name	Address	ProcessDate
1	Peter	1 Blue Rd.	5/1/2025
2	Samarth	22 Front St	5/1/2025



*Update the **customers** table with the **new customer** information*

updates (*source table with updates*)

CustomerID	Name	Address	Operation	ProcessDate
1	Peter	**99 Sunny St..**	**UPDATE**	5/15/2025
1	Peter	**123 Main St.**	**UPDATE**	5/20/2025
2	null	null	**DELETE**	5/20/2025
3	Kostas	100 Athens Dr.	**NEW**	5/20/2025

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Change Data Capture (CDC) Overview

Slowing Changing Dimensions (SCD) Type 1 (2 of 2)

customers (*target table with all customers*)

Update the row

Delete the row

Insert the row

CustomerID	Name	Address	ProcessDate
1	Peter	**1 Blue Rd.**	5/1/2025
2	Samarth	22 Front St	5/1/2025
3	***Kostas***	***100 Athens Dr.***	***5/20/2025***

updates (*source table with updates*)

CustomerID	Name	Address	Operation	ProcessDate
1	Peter	**99 Sunny St..**	**UPDATE**	**5/15/2025**
1	Peter	**123 Main St.**	**UPDATE**	**5/20/2025**
2	null	null	**DELETE**	5/20/2025
3	Kostas	100 Athens Dr.	**NEW**	5/20/2025

Updated SCD Type 1 Table

CustomerID	Name	Address	ProcessDate
1	Peter	**123 Main St.**	5/20/2025
3	Kostas	100 Athens Dr.	5/20/2025

- **Peter** was updated with the **latest** information
- **Samarth** was deleted from the table
- **Kostas** was added to the table

Used **SCD Type 1** when only the **current data matters** and historical values are irrelevant

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Change Data Capture (CDC) Overview

Slowing Changing Dimensions (SCD) Type 2 (*Outside the scope of the*

SCD Type 2 (Historical Tracking/Versioning)

- *When a record changes:*
  - The **old record is preserved** with an additional column indicating its validity period (start date, end date, or a current flag) and a **new record is inserted** with the updated information.
  - When a **record is marked as deleted,** the record is kept and a column indicates the record is **inactive**.
- Used when **historical data is important**, and the system needs to track **how attributes change over time** (like tracking changes in a customer's address or status).

Example customers target table with SCD Type 2

Columns indicate **active** and **inactive** rows

CustomerID	Name	Address	ProcessDate	__START_AT	__END_AT
1	Peter	123 Main St.	5/20/2025	5/20/2025	null
1	Peter	1 Blue Rd.	5/1/2025	5/1/2025	5/20/2025
2	Samarth	22 Front St	5/1/2025	5/1/2025	5/20/2025
3	Kostas	100 Athens Dr.	5/20/2025	5/20/2025	null

Null values indicate the **current row**

Non null values indicate **inactive (historic)** rows

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Change Data Capture (CDC) Overview

Using AUTO CDC INTO in Spark Declarative Pipelines (Formerly APPLY CHANGES INTO)

customers (*target table with all customers*)

CustomerID	Name	Address	ProcessDate
1	Peter	1 Blue Rd.	5/1/2025
2	Samarth	22 Front St	5/1/2025



updates (source table with updates)

CustomerID	Name	Address	Operation	ProcessDate
1	Peter	99 Sunny St..	**UPDATE**	5/15/2025
1	Peter	123 Main St.	**UPDATE**	5/20/2025
2	Samarth	22 Front St	**DELETE**	5/20/2025
3	Kostas	100 Athens Dr.	**NEW**	5/20/2025

```
CREATE OR REFRESH STREAMING TABLE customers;

CREATE FLOW scd_type_1_flow AS
AUTO CDC INTO customers
 FROM STREAM updates
 KEYS (CustomerID)
 APPLY AS DELETE WHEN operation = "DELETE"
 SEQUENCE BY ProcessDate
 COLUMNS * EXCEPT (operation)
 STORED AS SCD TYPE 1;
```

- Create a streaming table, then use AUTO CDC to populate it
- Apply updates, inserts and deletes to the **target** table
- Source records to determine **updates, deletes and inserts**
- The column(s) that **uniquely identify a row** in the source and target tables
- Specifies when a CDC event should be treated as a **DELETE** rather than an upsert.
- Specifies the **logical order of CDC events** in the source data
- Specifies a **subset of columns** to include in the **target**
- Whether to store records as **SCD type 1 (default) or type 2**

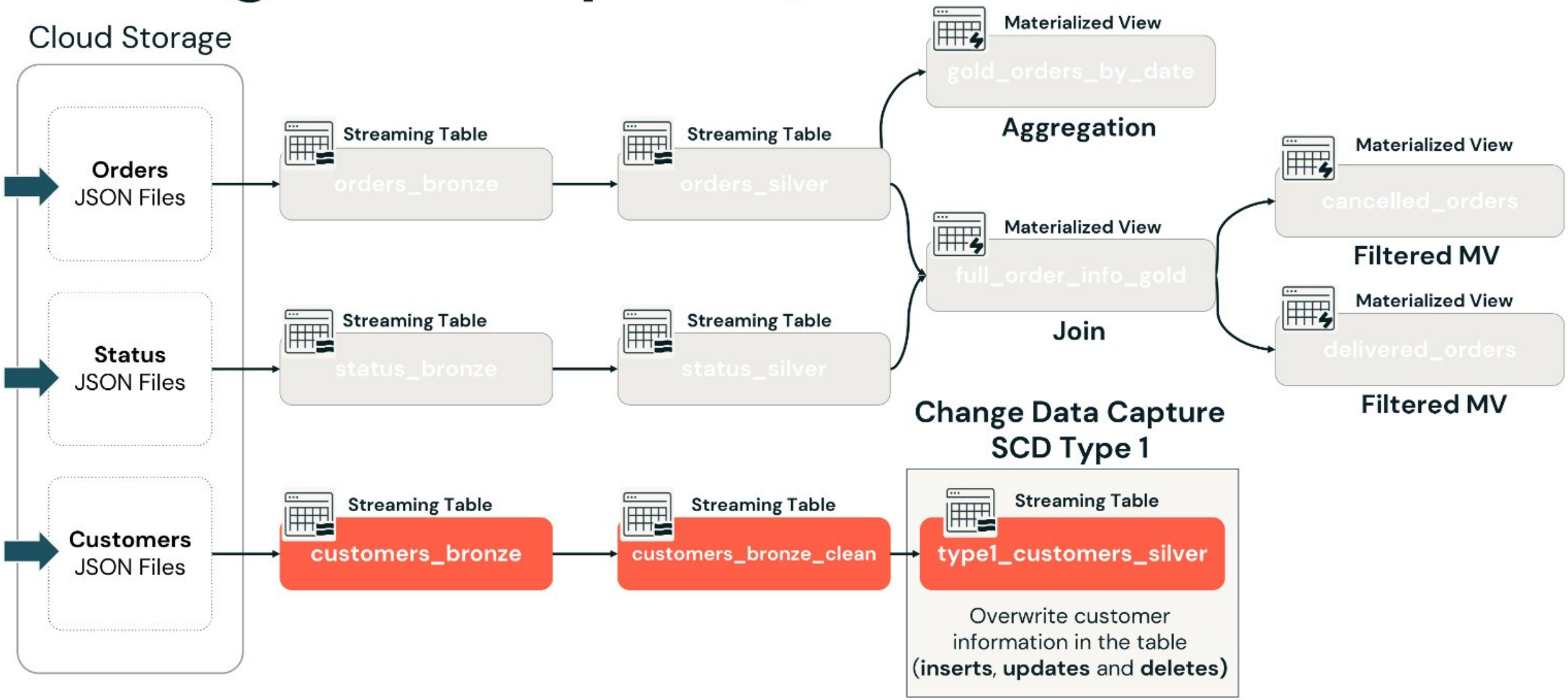
*INSERTs and UPDATEs are implicitly implemented using the **KEY(s)**, no coding required*

Use **AUTO CDC INTO (Formerly APPLY CHANGE INTO)** instead of a traditional batch job with complex **MERGE INTO** syntax

Use AUTO CDC FROM SNAPSHOT (Public Preview as of 2025Q2) to process changes in database snapshots.

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Change Data Capture (CDC) Overview



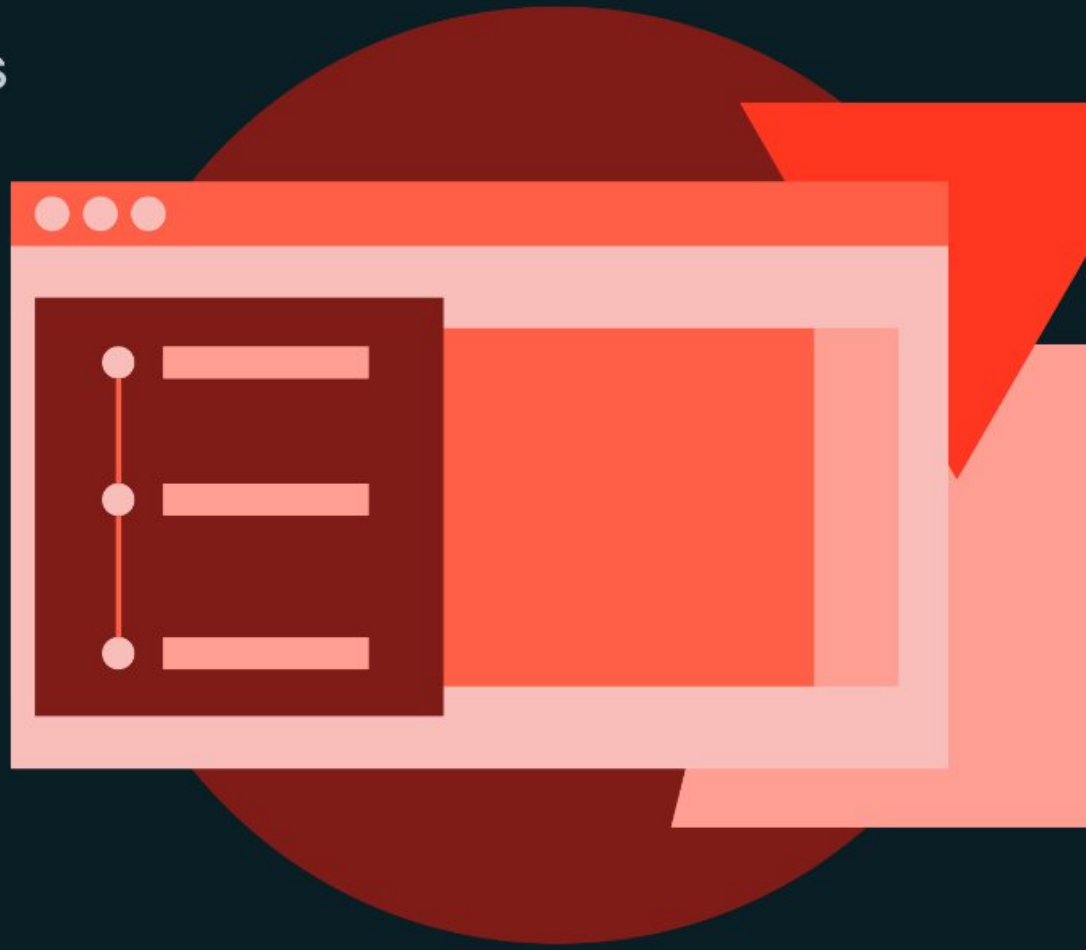
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Building Lakeflow Spark Declarative Pipelines

DEMONSTRATION

Change Data Capture with AUTO CDC INTO INTO



Notebook: 6 Demo - Change Data Capture with AUTO CDC with SCD TYPE 1

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LECTURE

Summary and Next Steps

Open **Notebook 7 – Summary and Next Steps** to explore additional topics and recommended next steps beyond this course.



Notebook: 7 - Summary and Next Steps

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Building Lakeflow Spark Declarative Pipelines

LAB EXERCISE

BONUS - AUTO CDC INTO with SCD Type 1

If you are interested in performing CDC with Spark Declarative Pipelines we have provided an extra lab you can complete after class.



Notebook: 8 Lab - AUTO CDC INTO with SCD Type 1

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